28.01.2022

# ALGEBRĂ

## Sesiune

1) Valori și vectori proprii. Diagonalizarea matricilor și endomorfismelor!

- $\lambda$ - valoare proprie
- $P(\lambda) = \det(A - \lambda I_n)$ - polinom caracteristic
- $\sigma(A)$ - spectrul $= \{\lambda_1, \lambda_2, \dots, \lambda_m\}$ val proprii
- mult algebrică - (de câte ori apare) la o putere este $\lambda$
- mult geometrică $\forall \lambda_0 \overset{not}{=} \dim_K V_{\lambda_0}$

Ex Să se det vectorii și valorile proprii pt $A = \begin{pmatrix} 1 & 0 \\ 8 & 2 \end{pmatrix}$

Pasi

1) Polinomul caract al A

$$P_A(\lambda) = \det(A - \lambda I_2) = \begin{vmatrix} 1-\lambda & 0 \\ 8 & 2-\lambda \end{vmatrix} = (1-\lambda)(2-\lambda)$$

2) Rădăcinile ecuației caracteristice $P_A(\lambda) = 0$.

$(1-\lambda)(2-\lambda) = 0 \Rightarrow \begin{cases} \lambda = 1 \\ \lambda = 2 \end{cases}$ (val proprii)

anume $\sigma(A) = \{1, 2\}$

3) Pt fiecare val proprie găsesc vectorii proprii asociați

Fie $\boxed{\lambda_1 = 1}$ Fie $v \in (a, b) \in \mathbb{R}^2$ vector propriu

$$\Rightarrow \boxed{A v^t = \lambda_1 v^t} \Rightarrow \begin{pmatrix} 1 & 0 \\ 8 & 2 \end{pmatrix} \cdot \begin{pmatrix} a \\ b \end{pmatrix} = 1 \begin{pmatrix} a \\ b \end{pmatrix}$$

$$\Rightarrow \begin{cases} a = a \\ 8a + 2b = b \end{cases} \Rightarrow 8a + b = 0$$

Fie $a = \alpha \Rightarrow b = -8\alpha$

$\Rightarrow v \in \{(\alpha, -8\alpha), \alpha \in \mathbb{R}\} = V_{\lambda_1}$ subspațiul propriu asociat val propriu $\lambda_1$

$\boxed{\lambda_2 = 2}$ și $v \in (a, b) \in \mathbb{R}^2$ vector propriu asociat

$$A v^t = \lambda_2 v^t \Rightarrow \begin{pmatrix} 1 & 0 \\ 8 & 2 \end{pmatrix} \begin{pmatrix} a \\ b \end{pmatrix} = 2 \begin{pmatrix} a \\ b \end{pmatrix}$$

$$\Rightarrow \begin{cases} a = 2a \Rightarrow a = 0 \\ 8a + 2b = 2b \end{cases} \Rightarrow b = \alpha \in \mathbb{R}$$

$\Rightarrow v \in \{(0, \alpha), \alpha \in \mathbb{R}\} = V_{\lambda_2}$ subs propriu asociat val propriu $\lambda_2$

ex SEMINAR pe metoda YT (ac acasă)

$$A = \begin{pmatrix} -1 & 3 & -1 \\ -3 & 5 & -1 \\ -3 & 3 & 1 \end{pmatrix}$$

$$P_A(\lambda) = \det(A - \lambda I_2) = \begin{vmatrix} -1-\lambda & 3 & -1 \\ -3 & 5-\lambda & -1 \\ -3 & 3 & 1-\lambda \end{vmatrix} \overset{C_1 \to C_1 + C_2}{=}$$

$$= \begin{vmatrix} 2-\lambda & 3 & -1 \\ 2-\lambda & 5-\lambda & -1 \\ 0 & 3 & 1-\lambda \end{vmatrix} = (2-\lambda) \begin{vmatrix} 1 & 3 & -1 \\ 1 & 5-\lambda & -1 \\ 0 & 3 & 1-\lambda \end{vmatrix}$$

$$= (2-\lambda) \begin{vmatrix} 1 & 3 & -1 \\ 0 & 2-\lambda & 0 \\ 0 & 3 & 1-\lambda \end{vmatrix} = (2-\lambda)^2 (1-\lambda)$$

$(2-\lambda)^2 (1-\lambda) = 0 \Leftrightarrow \lambda = 2$ cu multp 2

$\lambda = 1$ cu multp 1.

$V(A) = \{\lambda_1, 2\}$

**$\lambda_1 = 1$** $A v^t = \lambda_1 v^t$, Fie $v \in (a,b,c) \in \mathbb{R}^3$ vector propriu

$$\Rightarrow \begin{pmatrix} -1 & 3 & -1 \\ -3 & 5 & -1 \\ -3 & 3 & 1 \end{pmatrix} \begin{pmatrix} a \\ b \\ c \end{pmatrix} = 1 \begin{pmatrix} a \\ b \\ c \end{pmatrix}$$

$$\Rightarrow \begin{cases} -a + 3b - c = a \\ -3a + 5b - c = b \\ -3a + 3b + c = c \end{cases} \Rightarrow \begin{cases} -2a + 3b - c = 0 \\ -3a + 4b - c = 0 \\ -3(a - b) = 0 \Leftrightarrow -3a + 3b = 0 \end{cases}$$

$-2a + 3b - c = 0$

$3a - 3b = 0$ (+)

$\boxed{a - c = 0}$

$-3a + 4b - c = 0$

$3a - 3b = 0$

$\boxed{b - c = 0}$ $\Rightarrow \boxed{b = c}$

$+2a - 3b + c = 0$

$-3a + 4b - c = 0$

$-a + b = 0$ (+)

$$\begin{cases} a - c = 0 \\ b - a = 0 \\ b - c = 0 \end{cases} \Rightarrow \boxed{a = c}$$

Fie $a = \alpha \in \mathbb{R} \Rightarrow a = b = c = \alpha$

$\Rightarrow v \in \{(\alpha, \alpha, \alpha), \alpha \in \mathbb{R}\}$

$= \{\alpha(1,1,1), \alpha \in \mathbb{R}\} = V_{\lambda_1}$ subsp propriu asociat val propriu $\lambda_1 = 1$

**$\lambda_2 = 2$** Fie $v \in (a,b,c) \in \mathbb{R}^3$ vector propriu

$A v^t = 2 v^t$

$$\begin{pmatrix} -1 & 3 & -1 \\ -3 & 5 & -1 \\ -3 & 3 & 1 \end{pmatrix} \begin{pmatrix} a \\ b \\ c \end{pmatrix} = 2 \begin{pmatrix} a \\ b \\ c \end{pmatrix}$$

$$\Rightarrow \begin{cases} -a + 3b - c = 2a \\ -3a + 5b - c = 2b \\ -3a + 3b + c = 2c \end{cases} \Rightarrow \begin{cases} -3a + 3b - c = 0 \\ -3a + 3b - c = 0 \\ -3a + 3b - c = 0 \end{cases}$$

Fie $A - 2\mathcal{I}_3 = \begin{pmatrix} -3 & 3 & -1 \\ -3 & 3 & -1 \\ -3 & 3 & -1 \end{pmatrix} \to \begin{pmatrix} -3 & 3 & -1 \\ 0 & 0 & 0 \\ 0 & 0 & 0 \end{pmatrix}$

$-3a + 3b - c = 0 \Leftrightarrow \boxed{c = -3a + 3b}$

Fie $a = \alpha$ și $b = \beta$

$v \in \{(\alpha, \beta, -3\alpha + 3\beta), \alpha, \beta \in \mathbb{R}\} =$

$= \{\alpha \underbrace{(1,0,-3)}_{v_1} + \beta \underbrace{(0,1,3)}_{v_2}, \alpha, \beta \in \mathbb{R}\} = V_{\lambda_2}$ subsp propriu asociat val propriu $\lambda_2 = 2$

**Ex 2.** Să se afle vectorii și valorile proprii ale lui $f$:

$f: \mathbb{R}^2 \to \mathbb{R}^2$, $f(a,b) = (2a - b, b - a)$

$f(1,0) = (2,-1)$

$f(0,1) = (-1,1)$ $\Rightarrow M_f^{\mathcal{B}_c} = \begin{pmatrix} f(1,0) & f(0,1) \\ 2 & -1 \\ -1 & 1 \end{pmatrix} = A$

$P_A(\lambda) = \det(A - \lambda \mathcal{I}_2) = \begin{vmatrix} 2-\lambda & -1 \\ -1 & 1-\lambda \end{vmatrix} = (2-\lambda)(1-\lambda) - 1 \Rightarrow$

$\Rightarrow \lambda^2 - 3\lambda + 1 = 0$

$\Delta = 9 - 4 = 5 \Rightarrow \lambda_{1,2} = \frac{3 \pm \sqrt{5}}{2}$

$\Rightarrow \lambda_1 = \frac{3 + \sqrt{5}}{2}$, $\lambda_2 = \frac{3 - \sqrt{5}}{2}$

Se caută vectorii proprii

Fie $\lambda_1 = \frac{3 + \sqrt{5}}{2}$ și $v \in \mathbb{R}^2 = (a,b)$ vector propriu asociat

$\Rightarrow A v^t = \lambda_1 v^t$

$$\begin{pmatrix} 2 & -1 \\ -1 & 1 \end{pmatrix} \begin{pmatrix} a \\ b \end{pmatrix} = \frac{3 + \sqrt{5}}{2} \begin{pmatrix} a \\ b \end{pmatrix}$$

... continuare

ex 3 Să se det vectori și val proprii pt

$f \in \mathbb{R}^3 \to \mathbb{R}^3$, $f(a,b,c) = (a-b+c, b, b-c)$

$f(1,0,0) = (1,0,0)$
$f(0,1,0) = (-1,1,1)$
$f(0,0,1) = (1,0,-1)$

$\Rightarrow M_{f}^{\mathcal{B}_c} = \begin{pmatrix} 1 & -1 & 1 \\ 0 & 1 & 0 \\ 0 & 1 & -1 \end{pmatrix} = A$

$P_A(\lambda) = \det(A - \lambda I_3) = \begin{vmatrix} 1-\lambda & -1 & 1 \\ 0 & 1+\lambda & 0 \\ 0 & 1 & -1-\lambda \end{vmatrix} = (1-\lambda)(1+\lambda)(-1-\lambda)$

$= -(1-\lambda)^2(1+\lambda)$

$\det = 0 \Leftrightarrow -(1-\lambda)^2(1+\lambda) = 0 \Leftrightarrow \begin{cases} \lambda_1 = -1 \text{ cu mult. } 1 \\ \lambda_2 = 1 \text{ cu mult. } 2 \end{cases}$

$\sigma(A) = \{1, -1\}$ - spectrul $\quad m_a(\lambda_1) = 2$

$m_a(\lambda_2) = 1$

Vect proprii

$\lambda_1 = 1$ val proprie și $v = (a,b,c) \in \mathbb{R}^3$ vect. proprii asociat

$\Rightarrow A v^t = \lambda_1 v^t$ (sau) $(A - \lambda_1 I_3) v^t = 0$

$\begin{pmatrix} 1 & -1 & 1 \\ 0 & 1 & 0 \\ 0 & 1 & -1 \end{pmatrix} \begin{pmatrix} a \\ b \\ c \end{pmatrix} = \begin{pmatrix} a \\ b \\ c \end{pmatrix} \Rightarrow \begin{cases} a-b+c = a \\ b = b \\ b-c = c \end{cases} \Rightarrow \begin{matrix} b = c \\ b = 2c \end{matrix}$

$b = c = 0$

Fie $a = \alpha$, $\alpha \in \mathbb{R}$

$\Rightarrow V_{\lambda_1} = \{(\alpha, 0, 0) \mid \alpha \in \mathbb{R}\} = \{\alpha(1,0,0) \mid \alpha \in \mathbb{R}\}$

$\dim V_{\lambda_1} = 1$ mult. geometrică a $\lambda_1$

$mg(\lambda_1) = 1$.

$\lambda = -1$ și $v \in (a,b,c) \in \mathbb{R}^3$ vect. proprii asociat

$A v^t = \lambda v^t$

$\begin{cases} a-b+c = -a \\ b = -b \\ b-c = -c \end{cases} \Rightarrow 2a+c = 0 \Rightarrow \boxed{b = 0}$

Fie $a = \alpha \Rightarrow c = -2\alpha$

$\Rightarrow V_{\lambda_2} = \{(\alpha, 0, -2\alpha), \alpha \in \mathbb{R}\} = \{\alpha(1,0,-2), \alpha \in \mathbb{R}\}$

$\dim V_{\lambda_2} = 1 = mg(\lambda_2)$

## DIAGONALIZARE

- mult. geom = mult alg.
- bază a lui $V_\lambda$ ($\exists$ bază a lui $\mathbb{R}^2$ cu vect. proprii)

ex Să se diagonalizeze matricea

$B = \begin{pmatrix} 1 & 1 \\ 4 & 1 \end{pmatrix}$

$P_B(\lambda) = \det(B - \lambda I_2) = \begin{vmatrix} 1-\lambda & 1 \\ 4 & 1-\lambda \end{vmatrix} = (1-\lambda)^2 - 4$

$P_B(\lambda) = 0 \Rightarrow (1-\lambda)^2 = 4 \Rightarrow \lambda_1 = -1 \Rightarrow$
$\lambda_2 = +3$

$\Rightarrow \sigma(B) = \{-1, 3\}$ cu $\boxed{\begin{matrix} m_a(\lambda_1) = 1 \\ m_a(\lambda_2) = 1 \end{matrix}}$

Fie $\lambda_1 = -1$ și $v \in (a,b) \in \mathbb{R}^2$ vect. pr. asociat

$\Rightarrow B v^t = \lambda_1 v^t$

$\begin{pmatrix} 1 & 4 \\ 1 & 1 \end{pmatrix} \begin{pmatrix} a \\ b \end{pmatrix} = \begin{pmatrix} -a \\ -b \end{pmatrix} \Rightarrow \begin{cases} a+b = -a \\ 4a+b = -b \end{cases}$

$\Rightarrow \begin{cases} 2a+b = 0 \\ 4a+2b = 0 \end{cases}$

Fie $a=\alpha \Rightarrow b=-2\alpha$, $\alpha\in\mathbb{R}$

$\Rightarrow V_{\lambda_1}=\{(\alpha,-2\alpha)\,|\,\alpha\in\mathbb{R}\}$

$\dim V_{\lambda_1}=mg(\lambda_1)=1$.

Bază a lui $V_{\lambda_1}=B_{\lambda_1}=\{(1,-2)\}$

Fie $\boxed{\lambda=3}$ $\rightarrow v\in(a,b)$ vect propriu asociat $\Rightarrow Bv=3v$

$$\Rightarrow \begin{cases} a+b=3a \\ 4a+b=3b \end{cases} \Leftrightarrow \begin{cases} -2a+b=0 \\ 4a-2b=0 \end{cases}$$

Fie $a=\alpha \Rightarrow b=2\alpha$, $\alpha\in\mathbb{R}$

$V_{\lambda_2}=\{(\alpha,2\alpha)\,|\,\alpha\in\mathbb{R}\}$

$\dim V_{\lambda_2}=mg(\lambda_2)=1$.

Bază a lui $V_{\lambda_2}=\{(1,2)\}=B_{\lambda_2}$

$ma(\lambda_1)=mg(\lambda_1)$, $ma(\lambda_2)\overset{\text{şi}}{=}mg(\lambda_2)$ $\Rightarrow$ Crit. de diag $\Rightarrow$ $B$ e diagonalizabilă!

**Def** $B'=B_{\lambda_1}\cup B_{\lambda_2}=\{\underbrace{(1,-2)}_{v_1},\underbrace{(1,2)}_{v_2}\}$ bază a lui $\mathbb{R}^2$

Fie $\mathcal{B}\xrightarrow{T}\mathcal{B}'$

$$T=\begin{pmatrix} v_1 & v_2 \\ 1 & 1 \\ -2 & 2 \end{pmatrix}$$

$$B\sim T^{-1}BT=\begin{pmatrix} -1 & 0 \\ 0 & 3 \end{pmatrix}=\text{diag}(-1,3)$$

(valorile proprii!!!)

**Def** $B$ are forma diagonală $\begin{pmatrix} -1 & 0 \\ 0 & 3 \end{pmatrix}$ în baza $B'$

(*) $B^{2020}=?$

$$B\sim T^{-1}BT=\begin{pmatrix} -1 & 0 \\ 0 & 3 \end{pmatrix}=D=\text{diag}(-1,3)$$

$$T^{-1}BT=D \Rightarrow$$

$$\Rightarrow \boxed{B=TDT^{-1}}$$

$$B^2=TDT^{-1}TDT^{-1}=TD^2T^{-1}$$

$$\boxed{B^n=TD^nT^{-1}} \Rightarrow B^{2020}=TD^{2020}T^{-1}$$

$$D=\text{diag}(-1,3) \Rightarrow D^n=\text{diag}((-1)^n,3^n)=\begin{pmatrix} (-1)^{2020} & 0 \\ 0 & 3^{2020} \end{pmatrix}$$

$$T=\begin{pmatrix} 1 & 1 \\ -2 & 2 \end{pmatrix} \Rightarrow T^*=\begin{pmatrix} 2 & -1 \\ +2 & 1 \end{pmatrix}$$

$\det T=2+2=4\neq 0 \Rightarrow T$ inversabilă

$\delta_{11}=(-1)^2\cdot 2=2$

$\delta_{12}=(-1)^3\cdot(-2)=+2$

$\delta_{21}=(-1)^3\cdot 1=-1$

$\delta_{22}=(-1)^4\cdot 1=1$

$$T^{-1}=\frac{1}{\det T}T^*=\frac{1}{4}\begin{pmatrix} 2 & -1 \\ +2 & 1 \end{pmatrix}=\begin{pmatrix} \frac{1}{2} & -\frac{1}{4} \\ +\frac{1}{2} & \frac{1}{4} \end{pmatrix}$$

$$B^{2020}=\begin{pmatrix} 1 & 1 \\ -2 & 2 \end{pmatrix}\begin{pmatrix} 1 & 0 \\ 0 & 3^{2020} \end{pmatrix}\begin{pmatrix} \frac{1}{2} & -\frac{1}{4} \\ \frac{1}{2} & \frac{1}{4} \end{pmatrix}$$

**DE REȚINUT - diagonalizare**

- aflu valori și vectori proprii
  - $\det(A - \lambda I_m)$ → pt fiecare $\lambda$ → $Av^t = \lambda v^t$ și aflu $\dim V_\lambda$ + bază.
- multiplicitate egală ⇒ se diagonalizează (Criteriu)
- diagonalizez matricea → $T = (v_1 | v_2 | \dots)$

$$A \sim T^{-1} A T = \begin{pmatrix} \text{val prop} & 0 \\ 0 & \text{val prop} \end{pmatrix} = D$$

$$\boxed{A = T D T^{-1}}$$

$$A^m = T D^m T^{-1}$$

$$\boxed{\dim V_\lambda = m - \text{rang}(A - \lambda I_m)}$$

## 2) Spații euclidiene

$\vec{v} = a\vec{i} + b\vec{j}$, $\vec{w} = c\vec{i} + d\vec{j}$

produs scalar $\vec{v}\,\vec{w} = ac + bd \in \mathbb{R}$

lungimea vect $v = \sqrt{\vec{v} \cdot \vec{v}}$

cos dintre 2 vect $\cos(\vec{v}, \vec{w}) = \frac{\vec{v}\,\vec{w}}{\|\vec{v}\| \|\vec{w}\|}$

Produs scalar pe $\mathbb{R}^n$

$\langle \cdot, \cdot \rangle \; \mathbb{R}^m \times \mathbb{R}^m \longrightarrow \mathbb{R}$

$x = (x_1, \dots, x_m)$

$y = (y_1, \dots, y_m)$ $\Rightarrow \langle x, y \rangle = \sum_{i=1}^{m} x_i y_i$

- Norma și lungimea $\|x\| = \sqrt{\langle x, x \rangle} = \sqrt{\sum_{i=1}^{m} x_i^2}$
- $\cos(\widehat{x,y}) = \frac{\langle x, y \rangle}{\|x\| \|y\|}$
- $\langle x, y \rangle = 0 \Rightarrow x \perp y$

$(\mathbb{R}^m, \langle \cdot, \cdot \rangle)$ - spațiu euclidian real

- Produs scalar pe $(M_m(\mathbb{R}) \simeq \mathbb{R}^{m^2})$

$$\left\langle \begin{pmatrix} a & b \\ c & d \end{pmatrix}, \begin{pmatrix} a' & b' \\ c' & d' \end{pmatrix} \right\rangle = aa' + bb' + cc' + dd'$$

- Produs scalar pe $(\mathbb{R}_m[X] \simeq \mathbb{R}^{m+1})$

ex $\mathbb{R}_2[X] \simeq \mathbb{R}^3$ $\quad p = a + bx + cx^2$, $q = d + ex + fx^2 \in \mathbb{R}_2[X]$

$(a + bx + cx^2) \longrightarrow (a, b, c)$

$\langle p, q \rangle = ad + be + cf$.

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ex $\quad R_3[X] \simeq \mathbb{R}^4$

$p = 1 - x^2 + x^3$, $q = x - x^3$

$\langle p, q \rangle = 1 \cdot 0 + 0 \cdot 1 + (-1) \cdot 0 + 1 \cdot (-1) = -1$

$\|p\| = \sqrt{\langle p, p \rangle} = \sqrt{1 + 1 + 1} = \sqrt{3}$

$\|q\| = \sqrt{\langle q, q \rangle} = \sqrt{1 + 1} = \sqrt{2}$

$\cos \widehat{\langle p, q \rangle} = \frac{\langle p, q \rangle}{\|p\| \|q\|} = \frac{-1}{\sqrt{3}\sqrt{2}} = \frac{-1}{\sqrt{6}} \in [-1, 1]$. ∴

Prop

- $\langle \alpha x, y \rangle = \alpha \langle x, y \rangle$

  $\langle x, \alpha y \rangle = \alpha \langle x, y \rangle$

- $\langle x+y, z \rangle = \langle x, z \rangle + \langle y, z \rangle$

  $\langle x, y+z \rangle = \langle x, y \rangle + \langle x, z \rangle$

- $\langle \alpha x + \beta y, z \rangle = \alpha \langle x, z \rangle + \beta \langle y, z \rangle$

## COMPLEMENT ORTOGONAL !!

Def Fie $V \subseteq \mathbb{R}^n$

$V^\perp = \{ x \in \mathbb{R}^n \mid x \perp V \Leftrightarrow x \perp v, \forall v \in V \}$

↑ cornul ortogonal al lui V

Fie $B = \{b_1, b_t\}$ bază a lui V

~~$v \in V \Rightarrow \exists \alpha_1, \alpha_t \in \mathbb{R}$~~

- $V^\perp = \{ x \in \mathbb{R}^n \mid x \perp B, B \text{ bază a lui } V \}$
- $V \oplus V^\perp \simeq \mathbb{R}^n$

  Grassmann $\Rightarrow \dim V + \dim V^\perp = n$

  $\Rightarrow \boxed{\dim V^\perp = n - \dim V}$

ex $V = \{ (x,y,z) \in \mathbb{R}^3 \mid 2x - y + z = 0 \}$

a) Să se afle $V^\perp$ ( v ortogonal)

R

$2x - y + z = 0$

Fie $x = \alpha$ și $z = \beta \in \mathbb{R}$ $\Big| \Rightarrow y = 2\alpha + \beta$

$\Rightarrow V = \{ (\alpha, 2\alpha+\beta, \beta) \mid \alpha, \beta \in \mathbb{R} \}$

$\Downarrow \dim V = 2$ și o bază a lui V este $\{(1,2,0), (0,1,1)\}$

$\dim V^\perp = 3 - 2 = 1 \Rightarrow V^\perp = Sp\{(a,b,c) \in \mathbb{R}^3\}$

$(a,b,c) \in V^\perp \Rightarrow \forall (a,b,c) \not\perp B \Leftrightarrow$

$\Leftrightarrow \langle (a,b,c), B \rangle = 0$

$\Rightarrow \begin{cases} \langle (a,b,c), (1,2,0) \rangle = 0 \\ \langle (a,b,c), (0,1,1) \rangle = 0 \end{cases}$

$\Rightarrow \begin{cases} a + 2b = 0 \Rightarrow a = -2b = 2c \\ b + c = 0 \Rightarrow b = -c \end{cases}$

Not $b = \alpha \Rightarrow c = -\alpha$ și $a = -2\alpha$

$(a,b,c) \in \{(-2\alpha, \alpha, -\alpha) \mid \alpha \in \mathbb{R}\}$

$\Rightarrow V^\perp = \{(-2\alpha, \alpha, -\alpha), \alpha \in \mathbb{R}$

o bază în $V^\perp$ este $\{(-2,1,-1)\}$

## Bază ortonormală (de vectori)

Procedeul Gram-Schmidt.

Algoritmul Gram-Sch.

IN: bază $B = \{B_1, \dots, B_4\}$ a lui $\mathbb{R}^4$

OUT: $C = \{c_1, \dots, c_4\}$ bază de vectori

$\begin{cases} \|c_i\| = 1 \\ \langle c_i, c_j \rangle = 0, \forall i \neq j \end{cases}$

Dacă doar $\langle c_i, c_j \rangle = 0, \forall i \neq j \Rightarrow$ BAZĂ ORTOGONALĂ

Dacă doar $\|c_i\| = 1 \Rightarrow$ bază NORMATĂ

$\Downarrow +$

ORTONORMALĂ

ex $B = \{b_1 = (-1,0,1), b_2 = (1,-1,0), b_3 = (1,1,2)\}$

Bază (?) a lui $\mathbb{R}^3$ → $\begin{vmatrix} -1 & 1 & 1 \\ 0 & -1 & 1 \\ 1 & 0 & 2 \end{vmatrix} \overset{?}{\neq} 0 \Rightarrow$ DA e bază

Obținem o bază ortonormată din B

I) Bază ortogonală $\{c_1, c_2, c_3\}$

II) Normăm $\{c_1, c_2, c_3\}$

PAS 1 $c_1 = (-1,0,1) = b_1$

copiem ca vectori

PAS 2 $c_2 = b_2 + \alpha c_1$, $\alpha \in \mathbb{R}$ a.î. $\langle c_2, c_1 \rangle = 0$.

↗ facem ceruta birotată (ofiu $\alpha$)

$\langle c_2, c_1 \rangle = 0$

$\Downarrow$

$\langle b_2 + \alpha c_1, c_1 \rangle = 0 \Leftrightarrow \langle b_2, c_1 \rangle + \alpha \langle c_1, c_1 \rangle = 0$

$\Leftrightarrow$

$\langle b_2, c_1 \rangle = -1 + 0 + 0 = -1$

$\langle c_1, c_1 \rangle = 1 + 1 = 2$

$\Leftrightarrow \quad -1 + 2\alpha = 0 \Rightarrow \alpha = \frac{1}{2}$

$\Rightarrow c_2 = b_2 + \alpha c_1 = (1,-1,0) + \left(-\frac{1}{2}, 0, \frac{1}{2}\right) =$

$= \left(+\frac{1}{2}, -1, \frac{1}{2}\right)$

PAS 3 $c_3 = b_3 + \alpha c_1 + \beta c_2$ cu $\alpha, \beta \in \mathbb{R}$ și alți $\alpha$ a.î. $\langle c_3, c_1 \rangle = 0 = \langle c_3, c_2 \rangle$.

$\langle c_3, c_1 \rangle = \langle b_3 + \alpha c_1 + \beta c_2, c_1 \rangle = \langle b_3, c_1 \rangle + \alpha \langle c_1, c_1 \rangle$

$+ \beta \langle c_2, c_1 \rangle = 1 + \alpha\left(\frac{1}{2} \cdot 2\right) + \beta \underbrace{\left(-\frac{1}{2} + \frac{1}{2}\right)}_{0}$

$1 + 2\alpha = 0 \Rightarrow \alpha = -\frac{1}{2}$

$\langle c_3, c_2 \rangle = \langle b_3 + \alpha c_1 + \beta c_2, c_2 \rangle =$

$= \langle b_3, c_2 \rangle + \alpha \underbrace{\langle c_1, c_2 \rangle}_{0} + \beta \langle c_2, c_2 \rangle$

$= \frac{1}{2} + \beta \frac{3}{2} = 0$

$\Rightarrow \boxed{\beta = -\frac{1}{3}}$

$\Rightarrow c_3 = b_3 + \alpha c_1 + \beta c_2 = (1,1,2) - \frac{1}{2}(-1,0,1) - \frac{1}{3}\left(\frac{1}{2}, -1, \frac{1}{2}\right)$

$= (1,1,2) - \left(-\frac{1}{2}, 0, \frac{1}{2}\right) - \left(\frac{1}{6}, -\frac{1}{3}, \frac{1}{6}\right)$

$= \left(\frac{4}{3}, \frac{4}{3}, \frac{4}{3}\right)$

$\Rightarrow \{c_1, c_2, c_3\}$ bază ortogonală

II) Normare

ex $v = (1,2,3) \Rightarrow \|v\| = \sqrt{1+4+9} = \sqrt{14}$

COMENTARIU

$\left\| \frac{1}{\|v\|} (1,2,3) \right\| = 1$

$\Vert$

$\left\| \frac{1}{\sqrt{14}} (1,2,3) \right\| = 1 \Rightarrow \frac{1}{\|v\|} \cdot v$ NORMAT

$\frac{v}{\|v\|}$ nu există

$\|c_1\| = \sqrt{1+0+1} = \sqrt{2}$

$\|c_2\| = \sqrt{\left(\frac{1}{2}\right)^2 + 1 + \left(\frac{1}{2}\right)^2} = \sqrt{\frac{1}{4} + 1 + \frac{1}{4}} = \sqrt{\frac{3}{2}}$

$\|c_3\| = \sqrt{\left(\frac{4}{3}\right)^2 \cdot 3} = \sqrt{\frac{16}{9} \cdot 3} = \sqrt{\frac{16}{3}} = \frac{4\sqrt{3}}{3}$

fac invers

$\Rightarrow$

$\left\{ \frac{1}{\sqrt{2}} c_1, \sqrt{\frac{2}{3}} c_2, \frac{3}{4\sqrt{3}} c_3 \right\}$ bază ortonormată

Ex casa

Fie $V = \{(x_1,\dots,x_5) \in \mathbb{R}^5 \mid \begin{cases} x_1+x_2+x_3-x_4+x_5=0 \\ x_1+2x_2-x_4+x_5=0 \\ 3x_1+x_2+5x_3-3x_4+3x_5=0 \end{cases}\}$

fie $u=(1,1,3,-1,4)$

a) Să se determine complementul ortogonal ($V^\perp$) relativ la produsul scalar euclidian din $\mathbb{R}^5$ al spațiului pe $V$ a lui $u$ ($pr_V u$)

1) Aflu o bază a lui $V$.

$$\begin{cases} a+b+c-d+e=0 \\ a+2b-d+e=0 \\ 3a+b+5c-3d+3e=0 \end{cases} \Rightarrow \quad a+b+c = a+2b$$

$$\boxed{c=b}$$

$$2a = b-c \Rightarrow \boxed{a=0}$$

$$6b = 3d-3e \mid :3$$

$$2b = d-e \mid \Rightarrow \quad b = d-e \mid \Rightarrow$$

$$a+b+c = d-e$$

$$A \overset{mat}{=} \begin{pmatrix} 1&1&1&-1&1 \\ 1&2&0&-1&1 \\ 3&1&5&-3&3 \end{pmatrix} \overset{\substack{L_2\to L_2-L_1 \\ L_3\to L_3-3L_1}}{\longrightarrow} \begin{pmatrix} 1&1&1&-1&1 \\ 0&1&-1&0&0 \\ 0&-2&2&0&0 \end{pmatrix} \overset{L_3\to L_3+2L_2}{\longrightarrow}$$

$$\begin{pmatrix} 1&1&1&-1&1 \\ 0&1&-1&0&0 \\ 0&0&0&0&0 \end{pmatrix} \overset{L_1\to L_1-L_2}{\longrightarrow} \begin{pmatrix} x_1&x_2&x_3&x_4&x_5 \\ \hline 1&0&2&-1&1 \\ 0&1&-1&0&0 \\ 0&0&0&0&0 \end{pmatrix}$$

$$\begin{cases} x_1+2x_3-x_4+x_5=0 \\ x_2=x_3 \end{cases} \Rightarrow \begin{cases} x_1=-2x_3+x_4-x_5 \\ x_2=x_3 \end{cases}$$

$$\Rightarrow \begin{pmatrix} x_1 \\ x_2 \\ x_3 \\ x_4 \\ x_5 \end{pmatrix} = \begin{pmatrix} -2x_3-x_4-x_5 \\ x_3 \\ x_3 \\ x_4 \\ x_5 \end{pmatrix} \Rightarrow x_3 \underbrace{\begin{pmatrix} -2 \\ 1 \\ 1 \\ 0 \\ 0 \end{pmatrix}}_{v_1} + x_4 \underbrace{\begin{pmatrix} -1 \\ 0 \\ 0 \\ 1 \\ 0 \end{pmatrix}}_{v_2} + x_5 \underbrace{\begin{pmatrix} -1 \\ 0 \\ 0 \\ 0 \\ 1 \end{pmatrix}}_{v_3}$$

o bază pt $V$ este $\{v_1,v_2,v_3\}$

$\dim V^\perp = \dim \mathbb{R} - \dim V = 5-3 = 2$

(în continuare)

$$V^\perp = \{u \in \mathbb{R}^5 \mid \langle u, v_j \rangle = 0, \ j \in \{1,2,3\}\} = (a,b,c,d,e)$$

$$= (u_1,u_2,u_3,u_4,u_5)$$

$u = (1,1,3,-1,4)$ $\quad \{(u_1,u_2,u_3,u_4,u_5) \in \mathbb{R}^5 \mid \begin{cases} -2u_1+u_2+u_3=0 \\ -u_1+u_4=0 \\ -u_1+u_5=0 \end{cases}$

NV $\begin{cases} u_1+u_2 \\ \end{cases} \Rightarrow \begin{cases} -2u_1 = -u_2-u_3 \\ -u_1 = -u_4 \\ -u_1 = -u_5 \end{cases} \Rightarrow$

$$\Rightarrow \begin{cases} 2u_1-u_2 = u_3 \\ u_1 = u_4 \\ u_1 = u_5 \end{cases} \Rightarrow$$

$$\Rightarrow \{(u_1, u_2, 2u_1-u_2, u_1, u_1) \mid u_1,u_2 \in \mathbb{R}\}$$

$$\Rightarrow \{u_1 \underbrace{(1,0,2,1,1)}_{w_1} + u_2 \underbrace{(0,1,-1,0,0)}_{w_2}, \ u_1,u_2 \in \mathbb{R}^5\}$$

$\Rightarrow$ o bază pt $V^\perp$ este $\{w_1,w_2\}$

$u \in \mathbb{R}^5 \Rightarrow \underbrace{a w_1 + b w_2 + c w_3}_{pr_{V^\perp} u} + w = u. \quad (1)$ $\quad w \in V$

pt a det. $a,b,c$ vom înmulți scalar, pe rând relația (1) cu $w_1, w_2, w_3$

$$\langle u, w_1 \rangle = a\langle w_1,w_1\rangle + b\langle w_1,w_2\rangle + c\langle w_1,w_3\rangle + \underbrace{\langle w, w_1\rangle}_{0 \ (ortog)}$$

$$\langle u, w_2 \rangle = a\langle w_1,w_2\rangle + b\langle w_2,w_2\rangle + c\langle w_3,w_2\rangle + 0$$

$$\langle u, w_3 \rangle = a\langle w_1,w_3\rangle + b\langle w_2,w_2\rangle + c\langle w_3,w_3\rangle + 0$$

$$\begin{cases} (4+2)a + 2b + 2c = -2+1+3 \\ 6a+2b-c=0 \\ 2a-b+2c=3 \end{cases} \quad 6a+2b+2c=2$$

$$\begin{cases} 3a - b + c = 1 \\ 6a + 2b - c = 0 \\ 2a - b + 2c = 3 \end{cases}$$

$3a + b = 1 \mid (-3) \Rightarrow \boxed{b = 1}$

$14a + 3b = 3$

$13a = 0 \Rightarrow \boxed{a = 0}$

$\Rightarrow \boxed{c = 2}$

$pr_V u = a v_1 + b v_2 + c v_3 =$

$= v_2 + 2v_3 = (1,0,0,1,0) + (-2,0,0,0,2)$

$= (-1,0,0,1,2)$.

**De reținut!**

- prop produsului scalar
- complementul ortogonal
  - o bază pt $V$
  - scad și o bază pt $V^{\perp}$ — dim $V^{\perp}$
  - iau un vect de forma gen pe care îl înmulțesc scalar cu vect din baza $V$ (
  - fac sistem
- bază [ ortogonală → Gram-Schmidt !! { - copiez primul v; - fac comb liniară; a ? să am produs scalar 0; - aflu pe rând
  - calc normele pt vect care fac bază ortogonală
  - fac „inversa" (nu știu cum se zice)

↓

bază ortonormală

## 3) CONICE

Ex Aduceți la forma canonică Hessian și denumiți conica

$g(x,y) = 3x^2 - 4xy - 2x + 4y - 3 = 0$

**Metoda matricială** (val proprii)

Extrag forma pătratică asociată (extrag numai o dată, gradul total 2)

$g(x,y) = 3x^{\textcircled{2}} - 4xy^{\textcircled{1}}$

Scriu matricea asociată lui $g$

$$A = \begin{pmatrix} 3 & -2 \\ -2 & 0 \end{pmatrix}$$

coef lui $x^2$; coef lui $xy$; coef lui $yx$; coef lui $y^2$

★ Calc vectori și valorile proprii

$P_A(\lambda) = \det(A - \lambda I_2) = \begin{vmatrix} 3-\lambda & -2 \\ -2 & -\lambda \end{vmatrix} = (3-\lambda)(-\lambda) - 4 =$

$= -3\lambda + \lambda^2 - 4 =$

$= \lambda^2 - 3\lambda - 4$

$\Delta = 9 - 4 \cdot (-4) = 25$

$\lambda_{1,2} = \frac{3 \pm 5}{2}$ $\quad \lambda_2 = 4$, $\lambda_1 = -1$ — val proprii

$g = -1$ (gen) $\Rightarrow$ hiperbolă

Vectori proprii asociați

Fie $\lambda_1 = -1$ și $v = (a,b) \in \mathbb{R}^2$ vector propriu asociat

$A v^t = \lambda_1 v^t$

$$\begin{pmatrix} 3 & -2 \\ -2 & 0 \end{pmatrix} \begin{pmatrix} a \\ b \end{pmatrix} = \begin{pmatrix} -a \\ -b \end{pmatrix} \Rightarrow$$

$$\Rightarrow \begin{cases} 3a - 2b = -a \\ -2a = -b \end{cases} \Rightarrow 2a = b$$

Fie ~~$a=\alpha$~~ $\Rightarrow b=2\alpha \in \mathbb{R}$

$\Rightarrow V_{\lambda_1}=\{(\alpha,2\alpha),\ \alpha\in\mathbb{R}\}$ și o bază $B_{\lambda_1}=\{(1,2)\}$

Fie $\boxed{\lambda_2=4}$ $\Rightarrow v\in(a,b)\in\mathbb{R}^2$ vector propriu asociat

$$Av^t=\lambda_2 v^t \Rightarrow \begin{cases} 3a-2b=4a \\ -2a=4b \Rightarrow -a=2b \Rightarrow a=-2b \end{cases}$$

Fie $b=\alpha \Rightarrow a=-2\alpha, \alpha\in\mathbb{R}$

$\Rightarrow V_{\lambda_2}=\{(-2\alpha,\alpha),\ \alpha\in\mathbb{R}\}$ și o bază $B_{\lambda_2}=\{(-2,1)\}$

$B=B_{\lambda_1}\cup B_{\lambda_2}=\{(1,2),(-2,1)\}$ bază în $\mathbb{R}^2$

B este ortogonală !

$$\langle (1,2),(-2,1)\rangle = 0 \ !$$

**Normăm!**

$$\|(1,2)\| = \|(-2,1)\| = \sqrt{5}$$

$\Rightarrow B=\left\{\frac{1}{\sqrt{5}}(1,2),\ \frac{1}{\sqrt{5}}(-2,1)\right\}$ bază de versori (ortonormată)

**Matrica de rotație**

$$R=\begin{pmatrix} \frac{1}{\sqrt{5}} & \frac{-2}{\sqrt{5}} \\ \frac{2}{\sqrt{5}} & \frac{1}{\sqrt{5}} \end{pmatrix} \qquad \boxed{\det R=\pm 1}$$

Dacă $\det R=-1$ (inversează versorii). (Înmulțește a 2 col cu $-1$)

La noi, $\det R=1$

**Aplic rotația**

coord inițiale $\begin{pmatrix} x \\ y \end{pmatrix} = R \begin{pmatrix} x_1 \\ y_1 \end{pmatrix}$ coord rotite

$$\begin{cases} x=\frac{1}{\sqrt{5}}x_1-\frac{2}{\sqrt{5}}y_1 \\ y=\frac{2}{\sqrt{5}}x_1+\frac{1}{\sqrt{5}}y_1 \end{cases}$$

**Reper canonic**

$g(x,y) \longrightarrow g(x_1,y_1)$

$$3x^2-4xy-2x+4y-3=0$$

$$3\left(\frac{1}{\sqrt{5}}x_1-\frac{2}{\sqrt{5}}y_1\right)^2-4\left(\frac{1}{\sqrt{5}}x_1-\frac{2}{\sqrt{5}}y_1\right)\left(\frac{2}{\sqrt{5}}x_1+\frac{1}{\sqrt{5}}y_1\right)-2\left(\frac{1}{\sqrt{5}}x_1-\frac{2}{\sqrt{5}}y_1\right)-$$
$$-4\left(\frac{2}{\sqrt{5}}x_1+\frac{1}{\sqrt{5}}y_1\right)-3=$$

$$=3\left(\frac{1}{5}x_1^2+\frac{4}{5}y_1^2+\frac{4}{5}x_1y_1\right)-4\left(\frac{2}{5}x_1^2+\frac{1}{5}x_1y_1-\frac{4}{5}x_1y_1-\frac{2}{5}y_1^2\right)$$
$$-2\left(\frac{1}{\sqrt{5}}x_1-\frac{2}{\sqrt{5}}y_1\right)-4\left(\frac{2}{\sqrt{5}}x_1+\frac{1}{\sqrt{5}}y_1\right)-3=$$

$$=x_1^2\left(\frac{3}{5}-\frac{8}{5}\right)+y_1^2\left(\frac{12}{5}+\frac{8}{5}\right)+x_1y_1\left(-\frac{6}{5}+\frac{12}{5}\right)+x_1\left(-\frac{2}{\sqrt{5}}+\frac{8}{\sqrt{5}}\right)+y_1\left(\frac{4}{\sqrt{5}}+\frac{4}{\sqrt{5}}\right)$$
$$-3=0$$

$$\Rightarrow \boxed{-x_1^2+4y_1^2+\frac{6}{\sqrt{5}}x_1+\frac{8}{\sqrt{5}}y_1-3=0}$$

$$-\left(x_1^2-\frac{6}{\sqrt{5}}x_1+\frac{9}{5}\right)+4\left(y_1^2+\frac{2}{\sqrt{5}}y_1+\frac{1}{5}\right)-2=0$$

$$\left(x_1-\frac{3}{\sqrt{5}}\right)^2+4\left(y_1+\frac{1}{\sqrt{5}}\right)^2=2$$

**Translația**

$$\begin{cases} x_1-\frac{3}{\sqrt{5}}=X \\ y_1+\frac{1}{\sqrt{5}}=Y \end{cases}$$ coord translat

$\Rightarrow -x^2 + 4y^2 = 2 \mid 2$

$\Rightarrow -\frac{x^2}{2} + \frac{y^2}{1/2} = 1$ **HIPERBOLA**

## Invarianți

1) Genul conicei $\delta = \lambda_1 \lambda_2 = -4 \Rightarrow$ conica e hiperbolă

2) Centrul conicei $\begin{cases} \frac{\partial f}{\partial x} = 0 \\ \frac{\partial f}{\partial y} = 0 \end{cases}$

$\frac{\partial f}{\partial x} = 6x - 4y - 2 = 0 \Rightarrow y = 1$

$\frac{\partial f}{\partial y} = -4x + 4 = 0 \Rightarrow x = 1$

$\Rightarrow C(1,1)$

**Obs** Parabola nu are centru

**Pt diagonală** · hiperbolă $-x^2 + y^2 = 1$ cu centrul $(1,1)$

↘ rotație de unghiul $\theta$

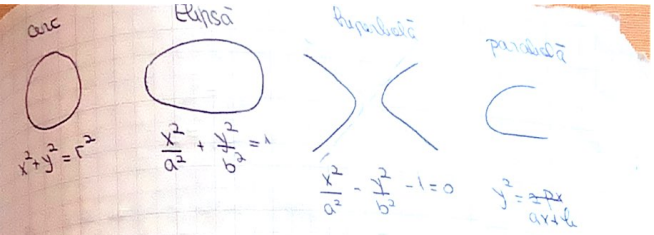
hiperbolă oarecare $\xrightarrow{\text{transl}}$ hip. funcție $-x^2 + \frac{y^2}{1/2} = 1$

$R = \begin{pmatrix} \frac{1}{\sqrt{3}} & \frac{-2}{\sqrt{3}} \\ \frac{2}{\sqrt{3}} & \frac{1}{\sqrt{3}} \end{pmatrix}$

$R_\theta = \begin{pmatrix} \cos\theta & -\sin\theta \\ \sin\theta & \cos\theta \end{pmatrix}$

$\cos\theta = \frac{1}{\sqrt{3}} < \frac{1}{2} \Rightarrow \theta > \frac{\pi}{3}$ sau $\theta < 2\pi - \frac{\pi}{3}$

$\sin\theta = \frac{2}{\sqrt{3}} >$




Gen $\begin{cases} \lambda_1 \lambda_2 > 0 - \text{elipsă} \\ \lambda_1 \lambda_2 < 0 - \text{hiperb} \\ \lambda_1 \lambda_2 = 0 - \text{parab} \end{cases}$

(ex) forma pătratică $f(x) = 3x_3^2 - 8x_1x_2 - 4x_1x_3 - 4x_2x_3$

$\Rightarrow A = \begin{pmatrix} 0 & -4 & -2 \\ -4 & 0 & -2 \\ -2 & -2 & 3 \end{pmatrix}$ coef $x_1^2$, coef $x_1x_2$, coef $x_1x_3$

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## Exercițiu curs

Folosind transformări ortogonale să se reducă la forma canonică forma pătratică

$f(x) = 3x_3^2 - 8x_1x_2 - 4x_1x_3 - 4x_2x_3$

$A = \begin{pmatrix} 0 & -4 & -2 \\ -4 & 0 & -2 \\ -2 & -2 & 3 \end{pmatrix}$

Determin $P(A)$

$\det(A - \lambda I_3) = \begin{vmatrix} -\lambda & -4 & -2 \\ -4 & -\lambda & -2 \\ -2 & -2 & 3-\lambda \end{vmatrix} \overset{C_1 \to C_1 - C_2}{=\!=} \begin{vmatrix} -\lambda+4 & -4 & -2 \\ -\lambda+4 & -\lambda & -2 \\ 0 & -2 & 3-\lambda \end{vmatrix}$

$= (4-\lambda) \begin{vmatrix} 1 & -4 & -2 \\ -1 & -\lambda & -2 \\ 0 & -2 & 3-\lambda \end{vmatrix} \overset{L_2 \to L_2 + L_1}{=\!=} (4-\lambda) \begin{vmatrix} 1 & -4 & -2 \\ 0 & -\lambda+4 & -4 \\ 0 & -2 & 3-\lambda \end{vmatrix}$

$= (4-\lambda) \cdot 1 \cdot (-1)^2 \begin{vmatrix} -\lambda+4 & -4 \\ -2 & 3-\lambda \end{vmatrix} = (4-\lambda)(\lambda^2 + \lambda - 20)$

$\lambda_1 = 4$

$\lambda_{2,3} = 4, -5$ $\Rightarrow$ $\sigma(A) = \{-5, 4\}$

cu $m_{\lambda=4} = 2$

$m_{\lambda=-5} = 1$

$\boxed{\lambda = -5}$ $\Rightarrow$ Caut $v \in (a,b,c)$ vector propriu $\in \mathbb{R}^3$ a.i.

$Av^t = \lambda v^t$

$$\begin{pmatrix} 0 & -4 & -2 \\ -4 & 0 & -2 \\ -2 & -2 & 3 \end{pmatrix} \begin{pmatrix} a \\ b \\ c \end{pmatrix} = \begin{pmatrix} -5a \\ -5b \\ -5c \end{pmatrix}$$

$$\Rightarrow \begin{cases} -4b - 2c = -5a \\ -4a - 2c = -5b \\ -2a - 2b + 3c = -5c \end{cases} \Rightarrow \begin{cases} 5a - 4b - 2c = 0 \\ 4a - 5b + 2c = 0 \\ 2a + 2b - 8c = 0 \end{cases}$$

$\Rightarrow$ $9a - 9b = 0 \Rightarrow a - b = 0 \Rightarrow \boxed{a = b}$

$4b - 8c = 0 \Rightarrow 4b = 8c \Rightarrow \boxed{b = 2c}$

Fie $c = \alpha \Rightarrow b = 2\alpha$ si $a = 2\alpha$, $\alpha \in \mathbb{R}$.

$\Rightarrow V_{\lambda=-5} = \{(2\alpha, 2\alpha, \alpha), \alpha \in \mathbb{R}\} =$

$= \{(2,2,1)\}$ $a_1$ baza pt $V_{\lambda=-5}$

$\boxed{\lambda = 4}$ Caut $v \in (a,b,c)$ vector propriu asociat

$Av^t = \lambda v^t$

$$\begin{cases} -4b - 2c = 4a \\ -4a - 2c = 4b \\ -2a - 2b + 3c = 4c \end{cases} \Rightarrow \begin{cases} 4a + 4b + 2c = 0 \\ 4a + 4b + 2c = 0 \\ 2a + 2b + c = 0 \end{cases} \Leftrightarrow \boxed{c = -2a - 2b}$$

$\Rightarrow$ Fie $a = \alpha$, $b = \beta$ $\Rightarrow c = -2\alpha - 2\beta$

$\Rightarrow V_{\lambda=4} = \{(\alpha, \beta, -2\alpha - 2\beta), \alpha, \beta \in \mathbb{R}\}$

$\Rightarrow \{(1,0,-2), (0,1,-2)\}$ baza pt $V_{\lambda=4}$

$a_2$ $a_3$

$\{\frac{1}{\|a_1\|} a_1\} = \{\frac{1}{3}\begin{pmatrix} 2 \\ 2 \\ 1 \end{pmatrix}\}$

(21) baza ON pt $V_{\lambda=-5}$

$\|a_1\| = \sqrt{2^2 + 2^2 + 1^2} = \sqrt{9} = 3$

$\{\frac{1}{\|a_2\|} \cdot a_2\} = \{\frac{1}{\sqrt{5}}\begin{pmatrix} 1 \\ 0 \\ -2 \end{pmatrix}\}$

$\{\frac{1}{\|a_3\|} a_3\} = \{\frac{1}{\sqrt{5}}\begin{pmatrix} 0 \\ 1 \\ -2 \end{pmatrix}\}$

• Aplic G-S baza $\{a_2, a_3\}$ pt a obt o baza ORTOGONALA a lui $V_{\lambda=4}$

pas 1 $v_1 = (1,0,-2) = a_2$

pas 2 Fac o combinatie liniara sau $v_2 = a_3 - \frac{\langle a_3, v_2\rangle}{\langle v_2, v_2\rangle} \cdot v_2$.

$v_2 = a_3 + \alpha a_2 = a_3 + \alpha v_1$ a.i. $\langle v_2, v_1\rangle = 0$

$\langle v_2, v_1\rangle = 0 \Leftrightarrow \langle a_3 + \alpha a_2, a_2\rangle = 0 \Leftrightarrow$

$\Leftrightarrow \langle a_3, a_2\rangle + \alpha\langle a_2, a_2\rangle = 0$

$\Leftrightarrow 6 + \alpha \cdot 5 = 0 \Rightarrow 5\alpha = -6 \Rightarrow \boxed{\alpha = -\frac{6}{5}}$

reluc calc $\begin{cases} v_2 = (0,1,-2) + (-\frac{6}{5}, 0, \frac{12}{5}) = (-\frac{6}{5}, 1, \frac{2}{5}) \\ \Rightarrow \{v_1, v_2\}$ o baza ortogonala

$\langle a_3, a_2\rangle + \alpha\langle a_2, a_2\rangle = 0$

$\Leftrightarrow 4 + 5\alpha = 0 \Rightarrow 5\alpha = -4 \Rightarrow \alpha = -\frac{4}{5}$

$v_2 = (0,1,-2) - \frac{4}{5}(1,0,-2) =$

$= \frac{1}{5}\begin{pmatrix} -4 \\ 5 \\ -2 \end{pmatrix}$.

$\|v_2\| = \sqrt{\langle v_2, v_2 \rangle} = \sqrt{5}$

$\|v_3\| = \frac{1}{5}\sqrt{16+25+4} = \frac{3\sqrt{5}}{5}$

$\Rightarrow$ o bază ORTONORMALĂ pt $V_{\lambda=4}$ este $\left\{\frac{1}{\|v_2\|} v_2 \times \frac{1}{\|v_3\|/v_3}\right\}$

$= \left\{\frac{1}{\sqrt{5}}\begin{pmatrix}1\\0\\-2\end{pmatrix}, \frac{1}{3\sqrt{5}}\begin{pmatrix}-4\\5\\-2\end{pmatrix}\right\}$

$q_2$ $q_3$

Pt $Q = (q_1|q_2|q_3)$ are loc $Q^T A Q = \begin{pmatrix} -5 & 0 & 0 \\ 0 & 4 & 0 \\ 0 & 0 & 4 \end{pmatrix}$ } val proprii

sau forma patratică $f$ are forma canonică

$f(y_1 q_1 + y_2 q_2 + y_3 q_3) = -5y_1^2 + 4y_2^2 + 4y_3^2$ relativ la baza ON $\{q_1, q_2, q_3\}$ a lui $\mathbb{R}^3 = \mathcal{M}_{3,1}(\mathbb{R})$

**DE REȚINUT** – Conica & forma canonică

Pt a det forma canonică

I) Metoda transformării ortogonale

- Scriu matricea asociată $\rightarrow$ „forma patratică"
- determin $\sigma(A)$ și câte o bază ORTONORMALĂ în fiecare subspațiu propriu $V_\lambda$
- $Q = (q_1|q_2|q_3)$, unde $q_1 = \left\{\frac{1}{\|v_1\|} v_1\right\}$ bază ON

  are loc $Q^T A Q = \begin{pmatrix} val_1 & 0 & 0 \\ 0 & val_2 & 0 \\ 0 & 0 & val_3 \end{pmatrix}$
- $f(y_1 q_1 + y_2 q_2 + y_3 q_3) = val_1 y_1^2 + val_2 y_2^2 + val_3 y_3^2$ relativ la baza ON $\{q_1, q_2, q_3\}$

II) Metoda valorilor și vectorilor proprii

- Scriu matricea A
- aflu $\sigma(A)$ și vectorii proprii
- normez $\Rightarrow R = \left(\frac{v_1}{\|v_1\|} \middle| \frac{v_2}{\|v_2\|}\right)$
- $\det R = \pm 1$
- aplic relația $\begin{pmatrix} x \\ y \end{pmatrix} = R \begin{pmatrix} x_1 \\ y_1 \end{pmatrix}$ coord. vect. / coord. mat. $\Rightarrow \begin{cases} x = \\ y = \end{cases}$
- reduc conica (înlocuiesc $x, y$ cu $x_1, y_1$)
- aduc la o formă din „tabel"
- Tabelul: general $\lambda_1 \cdot \lambda_2$; central $\left(\frac{\partial g}{\partial x} = 0, \frac{\partial g}{\partial y} = 0\right)$
- discuție?

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Ex Curs

a) Să se stabilească dacă endomorfismul $f: \mathbb{R}_3[X] \to \mathbb{R}_3[X]$ $f(a+bx+cx^2+dx^3) = a+d+bX+(c-2d)X^2+(a-2c+7d)X^3$ este diagonalizabil și în caz afirm să se diagonalizeze. (Să nu det o bază a sp vect $\mathbb{R}_3[X]$ relativ la care matricea asociată lui $f$ să fie o matrice diagonală).

$\mathcal{B} = \{1, X, X^2, X^3\}$

$f(1) = 1 + X^3$

$f(X) = X$

$f(X^2) = X^2 - 2X^3$

$f(X^3) = 1 - 2X^2 + 5X^3$

$\Rightarrow A = \begin{pmatrix} 1 & 0 & 0 & 1 \\ 0 & 1 & 0 & 0 \\ 0 & 0 & 1 & -2 \\ 1 & 0 & -2 & 5 \end{pmatrix}$

matricea asociată

$P_A(\lambda) = \det(A - \lambda I_4) = \begin{vmatrix} 1-\lambda & 0 & 0 & 1 \\ 0 & 1-\lambda & 0 & 0 \\ 0 & 0 & 1-\lambda & -2 \\ 1 & 0 & -2 & 5-\lambda \end{vmatrix} =$

$= (1-\lambda)^3 \begin{vmatrix} 1 & 0 & 0 & 1 \\ 0 & 1 & 0 & 0 \\ 0 & 0 & 1 & -2 \\ 1 & 0 & -2 & 5-\lambda \end{vmatrix}$ $L_4 \to L_4 - L_1$

$= (1-\lambda)^3 \begin{vmatrix} 1 & 0 & 0 & 1 \\ 0 & 1 & 0 & 0 \\ 0 & 0 & 1 & -2 \\ 0 & 0 & -2 & 4-\lambda \end{vmatrix}$ $L_4 \to L_4 + 2L_3$ $\begin{vmatrix} 1 & 0 & 0 & 1 \\ 0 & 1 & 0 & 0 \\ 0 & 0 & 1 & -2 \\ 0 & 0 & 0 & -\lambda \end{vmatrix} =$

~~$= (1-\lambda)^3 \cdot (-\lambda) = \lambda(\lambda-1)^3$~~

$= \lambda(\lambda-1)^2(\lambda-6)$

$\lambda = 1$ cu $m_{\lambda=1} = 2$

$\lambda = 0$, $m_{\lambda=0} = 1$

$\lambda = 6$, $m_{\lambda=6} = 1$

$\Rightarrow V(A) = \{0, 1, 6\}$

Cum $\forall A \in \mathcal{M}_{4\times4}(\mathbb{R})$ $\quad 1 \le r_\lambda \le m_\lambda$, $m_\lambda = 1 \Rightarrow r_\lambda = 1$

$\Rightarrow$ pt $\lambda \in \{0, 6\} \Rightarrow \boxed{r_\lambda = 1 = m_\lambda}$

Determinam $\boxed{r_{\lambda=1}} = \dim V_{\lambda=1}$.

$A - I_4 = \begin{pmatrix} 0 & 0 & 0 & 1 \\ 0 & 0 & 0 & 0 \\ 0 & 0 & 0 & -2 \\ 1 & 0 & -2 & 4 \end{pmatrix} \xrightarrow{L_1 \leftrightarrow L_4} \begin{pmatrix} 1 & 0 & -2 & 4 \\ 0 & 0 & 0 & 0 \\ 0 & 0 & 0 & -2 \\ 0 & 0 & 0 & 1 \end{pmatrix} \begin{matrix} L_4 \to L_4 \\ L_2 \leftrightarrow L_4 \end{matrix}$

$\begin{pmatrix} 1 & 0 & -2 & 4 \\ 0 & 0 & 0 & 1 \\ 0 & 0 & 0 & -2 \\ 0 & 0 & 0 & 0 \end{pmatrix} \xrightarrow{L_3 \to L_3 + 2L_4} \begin{pmatrix} \boxed{1} & 0 & -2 & 4 \\ 0 & 0 & 0 & \boxed{1} \\ 0 & 0 & 0 & 0 \\ 0 & 0 & 0 & 0 \end{pmatrix} \Rightarrow$

rang $= 2 \Rightarrow \dim_{\mathbb{R}} V_{\lambda=1} = 4$ (nr coll) $- 2 = 2$

$\Rightarrow r_{\lambda=1} = m_{\lambda=1} = 2$

$\Rightarrow$ A diag $\Leftrightarrow$ f diagonalizabilă

Sa continuam, aflu vectorii proprii (dat câte o bază pt. fiecare valoare proprie)

$\boxed{\lambda = 1}$ Caut $v \in (a, b, c, d) \in \mathbb{R}^4$

$Av^t = \lambda v^t$

$\begin{pmatrix} 1 & 0 & 0 & 1 \\ 0 & 1 & 0 & 0 \\ 0 & 0 & 1 & -2 \\ 1 & 0 & -2 & 5 \end{pmatrix} \begin{pmatrix} a \\ b \\ c \\ d \end{pmatrix} = 1 \begin{pmatrix} a \\ b \\ c \\ d \end{pmatrix}$

$\Rightarrow \begin{cases} a + d = a \\ b = b \\ c - 2d = c \\ a - 2c + 5d = d \end{cases} \Rightarrow \boxed{d = 0}$

$\Rightarrow a - 2c = 0 \Rightarrow \boxed{a = 2c} \Rightarrow c = \frac{a}{2}$

Fie $c = \alpha$ și $b = \beta$, $\alpha, \beta \in \mathbb{R}$

$\Rightarrow V_{\lambda=1} = \{(2\alpha, \beta, \alpha, 0), \alpha, \beta \in \mathbb{R}\}$

$= Sp\{(2, 0, 1, 0), (0, 1, 0, 0)\}$ ($v_1$, $v_2$)

$\boxed{\lambda = 0}$ Caut $v \in (a, b, c, d) \in \mathbb{R}^4$

$Av^t = \lambda v^t$

$\begin{cases} a + d = 0 \Rightarrow \boxed{a = -d} \\ b = 0 \\ c - 2d = 0 \Rightarrow \boxed{c = 2d} \\ a - 2c + 5d = 0 \end{cases}$ $\quad c - 2d = 0$, $4d - 2c = 0$

Fie $d = \alpha$, $\alpha \in \mathbb{R}$

$V_{\lambda=0} = \{(-\alpha, 0, 2\alpha, \alpha), \alpha \in \mathbb{R}\}$

$\Downarrow$ o bază este $(-1, 0, 2, 1)$ ($v_3$)

$\lambda=6 \Rightarrow$ Caut $v=(a,b,c,d)\in\mathbb{R}^4$

$$\begin{cases} a+d=6a \\ b=6b \Rightarrow b=0 \\ c-2d=6c \\ a-2c+5d=6d \end{cases} \Rightarrow \begin{cases} \boxed{d=5a} \\ b=0 \\ -2d=5c \Rightarrow c=\frac{-2d}{5}= \\ a-2c-d=0 \end{cases}$$

$a-2c-5a=0 \Rightarrow -2c-4a=0 \Rightarrow -2c=4a \Rightarrow \boxed{-c=2a}$

Fie $a=\alpha, \alpha\in\mathbb{R}$

$V_{\lambda=6}=\{(\alpha,0,-2\alpha,5\alpha)\,|\,\alpha\in\mathbb{R}\}$

• bază este $\{(1,0,-2,5)\}$
$v_4$

Fie $T=(v_1|v_2|v_3|v_4)$

$$\Downarrow$$

$$A\sim T^{-1}AT=\begin{pmatrix}1&0&0&0\\0&1&0&0\\0&0&0&0\\0&0&0&6\end{pmatrix}\overset{not}{=}D \Rightarrow$$

$\Rightarrow$ în baza $B=\{X, 2+X^2, -1+2X^2+X^3, 1-2X^2+5X^3\}$ a lui $\mathbb{R}_3[X]$, $f$ are matricea asociată $D$

Teoria Gram-Schmidt

$v_1=a_1$

$v_2 \overset{def}{=} a_2-\underbrace{\frac{\langle a_2,v_1\rangle}{\langle v_1,v_1\rangle}}_{pr_{v_1}a_2}\cdot v_1$

Fie $f_1:[-1,1]\to\mathbb{R}$, $f_1(x)=1$, $f_2(x)=x$, $f_3(x)=x^2$

$S=\{f_1,f_2,f_3\}\subseteq C_{[-1,1]}\overset{not}{=}\{f:[-1,1]\to\mathbb{R}\,|\,f \text{ cont}\}$

Să se det o bază ON pt $S$ relativ la $ps\ \langle f,g\rangle \overset{def}{=}\int_{-1}^{1} f(x)g(x)\,dx$ definit pe $C_{[-1,1]}$

1) ORTOGONALĂ

Fie $v_1=f_1=1$

$v_2=f_2-\frac{\langle f_2,p_1\rangle}{\langle p_1,p_1\rangle}\cdot p_1=x-\frac{\int_{-1}^{1}x\,dx}{\int_{-1}^{1}1\,dx}\cdot 1=$

$=x-\frac{\frac{x^2}{2}\Big|_{-1}^{1}}{x\Big|_{-1}^{1}}=x-\frac{\frac{1}{2}-\frac{1}{2}}{2}=x-0=x$

$p_3=f_3-\frac{\langle f_3,p_1\rangle}{\langle p_1,p_1\rangle}\cdot p_1-\frac{\langle f_3,p_2\rangle}{\langle p_2,p_2\rangle}p_2=$

$=x^2-\frac{\int_{-1}^{1}x^2dx}{\int_{-1}^{1}1\,dx}\cdot 1-\frac{\int_{-1}^{1}x^3dx}{\int_{-1}^{1}x^2dx}\cdot x=$

$=x^2-\frac{\frac{x^3}{3}\Big|_{-1}^{1}}{x\Big|_{-1}^{1}}-\frac{\left(\frac{x^4}{4}\Big|_{-1}^{1}=0\right)}{\frac{x^3}{3}\Big|_{-1}^{1}}\cdot x=$

$=x^2-\frac{\frac{2}{3}}{2}=x^2-\frac{1}{3}$

$\Rightarrow \{p_1,p_2,p_3\}$ bază ON pt $S$
$1 \quad x \quad x^2-\frac{1}{3}$

2) NORMARE

$$\begin{cases} \|p_1\|=1 \\ \|p_2\|=\sqrt{2} \\ \|p_3\|= \end{cases}$$

$\langle p_1,p_1\rangle=\int_{-1}^{1}dx=2 \Rightarrow \|p_1\|=\sqrt{2}$

$\langle p_2,p_2\rangle=\int_{-1}^{1}x^2dx=\frac{2}{3} \Rightarrow \|p_2\|=\sqrt{\frac{2}{3}}$

$\langle p_3,p_3\rangle=\int_{-1}^{1}\left(x^2-\frac{1}{3}\right)^2dx=\frac{8}{45} \Rightarrow \|p_3\|=\sqrt{\frac{8}{45}}$

O bază ON pt $S$ este $\left\{\frac{1}{\sqrt{2}}, \sqrt{\frac{3}{2}}x, \sqrt{\frac{45}{8}}\left(x^2-\frac{1}{3}\right)\right\}$.

# Ecuații diferențiale

## I) Ecuații cu variabile separabile :

$$\boxed{y'(x) = f(x)\,g(y)}, \text{ unde } y'(x) = \frac{dy}{dx}$$

ex 1) $(1+x^2)y' = -xy$ cu $y(0)=2$.

1) scot derivata $y' = \frac{-xy}{1+x^2} = \frac{dy}{dx}$

$\Updownarrow$

2) trec cu $e$ cu $y$ într-o parte cu $dx$ de $x$ în alta

$$\frac{-x}{1+x^2}\,dx = \frac{1}{y}\,dy \qquad \left(\frac{1+x^2}{1+x}\right)$$

$\Updownarrow$ Integrez

3) prelucrez cât a.î. să iasă $y$ în $\lg$ de o ct $k$.

$$\int \frac{dy}{y} = \int \frac{-2x}{2(1+x^2)}\,dx$$

$\Updownarrow$

$$\ln|y| = -\frac{1}{2}\ln(1+x^2) + c$$

5) înlocuiesc în $y(a)=b$.

$$\Rightarrow \ln|y| = \ln\frac{1}{\sqrt{1+x^2}} + \ln e^c$$

6) fac calc. $\Rightarrow \ln|y| = \ln\frac{e^c}{\sqrt{1+x^2}} \Rightarrow |y| = \frac{e^c}{\sqrt{1+x^2}} \Rightarrow y = \pm\frac{e^c}{\sqrt{1+x^2}}$ (4) $\quad e^c = k$

$$\Rightarrow y = \frac{k}{\sqrt{1+x^2}} \xrightarrow{y(0)=2} \boxed{k=2} \quad (5)$$

Deci $y(x) = \frac{2}{\sqrt{1+x^2}}$ (6)

2) $(1+x^2)y\,y' + x(1+y^2) = 0$ știind că $y(1)=2$

$\Updownarrow$

$$y' = \frac{-x(1+y^2)}{(1+x^2)y} = \frac{dy}{dx}$$

$\Updownarrow$

$$\frac{-x}{1+x^2}\,dx = \frac{y}{1+y^2}\,dy$$

$\Updownarrow \int$

$$\int \frac{-x}{1+x^2}\,dx = \int \frac{y}{1+y^2}\,dy$$

• arm ct pe „latura" cu „x"
• arm y în stg.

$\Updownarrow$ $\ln +c$

$$-\frac{1}{2}\ln(1+x^2) = \frac{1}{2}\ln(1+y^2) \;/\cdot 2$$

$\Updownarrow$

NU mai scriem

$$\ln\frac{1}{\sqrt{1+y^2}} = -\ln\frac{1}{\sqrt{1+x^2}} + \ln e^c$$

$\Updownarrow$

$$\ln\frac{1}{\sqrt{1+y^2}} =$$

$$\ln(1+y^2) = -\ln(1+x^2) + \ln c^2$$

$\Updownarrow$

$$\ln(1+y^2) = \ln\frac{c^2}{1+x^2}$$

$\Updownarrow$

$$1+y^2 = \frac{c^2}{1+x^2} \Rightarrow y = \pm\sqrt{\frac{c^2}{1+x^2}-1}$$

sol gen. $\Rightarrow c \in \mathbb{R}$

$y>0$

$$\Rightarrow y = +\sqrt{\frac{c^2}{1+x^2}-1} \Rightarrow 2 = \sqrt{\frac{c^2}{2}-1} \Rightarrow c^2 = 10$$

$y(1)=2$

Deci $y(x) = \sqrt{\frac{10}{1+x^2}-1}$ sol particulară

3) CURS $(1+x^2)y' - 2xy = x$ cu $y(0) = \frac{1}{2}$

$$y' = \frac{x+2xy}{1+x^2} = \frac{dy}{dx}$$

$$\Downarrow$$

$$y' = \frac{x(1+2y)}{1+x^2} = \frac{dy}{dx}$$

$$\Downarrow$$

$$\frac{x}{1+x^2}\,dx = \frac{1}{1+2y}\,dy$$

$$\Downarrow$$

$$\int \frac{(1+2y)'}{1+2y}\,dy = \int \frac{x}{1+x^2}\,dx$$

$$\Downarrow$$

$$\frac{1}{2}\ln(1+2y) = \frac{1}{2}\ln(1+x^2) + \ln c \;\Big|\cdot 2$$

$$\Downarrow$$

$$\ln(1+2y) = \ln(1+x^2) + \ln c^2$$

$$\Downarrow$$

$$\ln(1+2y) = \ln\left[c^2(1+x^2)\right]$$

$$\Downarrow$$

$$1+2y = c^2(1+x^2)$$

$$\Downarrow$$

$$y = \frac{c^2(1+x^2)-1}{2}$$

$$\Downarrow\; y(0) = \frac{1}{2}$$

$$\frac{c^2(1+0)-1}{2} = \frac{1}{2} \Rightarrow c^2 - 1 = 1 \Rightarrow c^2 = 2$$

Deci $y(x) = \frac{2(1+x^2)-1}{2} = \frac{2+2x^2-1}{2} = \frac{1+2x^2}{2}$

4) CURS $x^2(\sin y)\,y' + (\cos^2 y)\ln x = 0$ $\quad \sin y \neq 0,\ x > 0$

$$\Downarrow$$

$$y' = \frac{-(\cos^2 y)\ln x}{x^2 \sin y} = \frac{dy}{dx}$$

$$\Downarrow$$

$$-\frac{\sin y}{\cos^2 y}\,dy = \frac{\ln x}{x^2}\,dx$$

$$\Downarrow \int$$

$$+\int \frac{\sin y}{\cos^2 y}\,dy = -\int \frac{\ln x}{x^2}\,dx \quad (*)$$

$(\cos y)' = -\sin y \cdot y' = \sin y$

$$-\int \frac{\ln x}{x^2}\,dx = \int \left(\frac{1}{x}\right)'\ln x\,dx = \frac{\ln x}{x} - \int \frac{1}{x^2}\,dx = \frac{\ln x}{x} + \frac{1}{x} + c = \frac{1}{x}(1+\ln x) + C_1$$

$$\int \frac{\sin y}{\cos^2 y}\,dy = \int \frac{\sin y}{\sin^2 y}\,dy \overset{\sin y = t}{=} \int \frac{1}{t^2}\,dt = -\frac{1}{\sin y} + C_2$$

$$\int \frac{\sin y}{\cos^2 y}\,dy = \int \frac{(\cos y)'}{\cos^2 y}\,dy \overset{\cos y = t}{=} -\int \frac{1}{t^2}\,dt = \frac{1}{\cos y} + C_2$$

Revin la $(*)$ $\quad \frac{1}{\cos y} = \frac{1}{x}(1+\ln x) + C,\ C \in \mathbb{R}$

## II) Ecuații diferențiale omogene

TIP 1 $\boxed{y' = f\left(\frac{y}{x}\right)}$

(ex) $(x^2+y^2)\,y' = xy$, știind $y(2) = 1$.

1) Sct tavia. $y' = \frac{xy}{x^2+y^2} \,(x^2) = \frac{\frac{y}{x}}{1+\left(\frac{y}{x}\right)^2}$

2) Fac schimb de var.: $\frac{y}{x} = u \Rightarrow y = xu \Rightarrow y' = x'u + xu'$

$x' \cdot u + x \cdot u' = \frac{u}{1+u^2} \Rightarrow u' = \frac{\frac{u}{1+u^2} - u}{x} = \frac{u - u - u^3}{x}$

$\Rightarrow u' = \frac{-u^3}{x(1+u^2)}$ ecuație cu VS, $u' = \frac{du}{dx}$

$\Rightarrow \frac{du}{dx} = \frac{-u^3}{x(1+u^2)}$

3) Încercăm să rezolvăm o ec cu v.s.

$\Downarrow$

$\frac{1+u^2}{u^3} du = -\frac{1}{x} dx$

$\Downarrow \int$

$\frac{1}{-2} u^{-2} + \ln|u| = -\ln|x| + c$

$\Downarrow$

$\ln\left(e^{-\frac{1}{2u^2}} \cdot |u|\right) = \ln\left(\frac{e^c}{|x|}\right)$

$\Downarrow$

$e^{-\frac{1}{2u^2}} |u| = \frac{e^c}{|x|}$

$\Downarrow$

$e^{-\frac{1}{2u^2}} \cdot u = \pm \frac{e^c}{x} = K \in \mathbb{R}^*$

$e^{-\frac{1}{2u^2}} \cdot u = \frac{K}{x}$ $\quad u = \frac{y}{x}$ $\quad e^{-\frac{1}{2\frac{y^2}{x^2}}} \cdot \frac{y}{x} = \frac{K}{x} \mid \cdot x$

$\Rightarrow e^{-\frac{x^2}{2y^2}} y = K$ $\quad y(2)=1$ $\quad \boxed{K = e^{-\frac{1}{2}}}$

Deci $y(x) = e^{-\frac{x^2}{2y^2}} y = e^{-2}$

Ex 2 (CURS)

$y' = \frac{4x - 3y}{3x - 2y}$

$\Rightarrow y' = \frac{4 - 3\frac{y}{x}}{3 - 2\frac{y}{x}}$

La curs, lucrează cu z în loc de u.

$z = \frac{y}{x}$ $\Rightarrow y = x z \Rightarrow \boxed{y' = z + x z'}$

$\Rightarrow z + x z' = \frac{4 - 3z}{3 - 2z}$

$\Rightarrow x z' = \frac{4 - 3z}{3 - 2z} - z = \frac{4 - 3z - 3z + 2z^2}{3 - 2z}$

$\Rightarrow x z' = \frac{4 - 6z + 2z^2}{3 - 2z}$

$\Rightarrow z' = \frac{1}{x} \cdot \frac{2z^2 - 6z + 4}{3 - 2z}$ ec cu var sep

$\Delta = 36 - 4 \cdot 2 \cdot 4$

$36 - 32 = 4$

$z_{1,2} = \frac{6 \pm 2}{4}$ $\nearrow 2$ $\searrow 1$

$z' = \frac{1}{x} \cdot \frac{2(z-1)(z-2)}{3-2z}$

$\Downarrow$

$\frac{dz}{dx} = \frac{1}{x} \cdot \frac{2(z-1)(z-2)}{3-2z}$

$\Downarrow$

$\frac{3 - 2z}{(z-1)(z-2)} dz = \frac{2}{x} dx$

$\Downarrow \int$

$\int \frac{3 - 2z}{(z-1)(z-2)} dz = 2\int \frac{1}{x} dx + c$

$\int \frac{1}{x} = \ln|x|$

$\Downarrow$

$-\int \left(\frac{1}{z-1} + \frac{1}{z-2}\right) dz = \ln(x^2) + c$

$\Downarrow$

$-\ln|z-1| + \ln|z-2| = \ln x^2 + c$

$\Downarrow$

$-\ln\left|(z-1)(z-2)\right| = \ln x^2 + c$

$\Downarrow$

$(z-1)(z-2) \cdot x^2 = K \in \mathbb{R}$

$\Downarrow$

$\left(\frac{y}{x} - 1\right)\left(\frac{y}{x} - 2\right) x^2 = K$

$\Downarrow$

$(y - x)(y - 2x) = K, \; K \in \mathbb{R}$

„TIP 2"

$\boxed{y' = f(ax+by+c)}$ $a,b,c \in \mathbb{R}$

Schimbarea $z(x) = ax + by(x) + c$.

(ex) curs

$(x-y-1) + y'(y-x+2) = 0$

$$y' = -\frac{x-y-1}{y-x+2} = -\frac{x-y-1}{x-y-2}$$

$z(x) = x - y(x) - 1 \Rightarrow y(x) = -z(x) + x - 1$

$\Rightarrow y'(x) = 1 - z'(x)$

Înlocuiesc: $1 - z' = \frac{-x+y+1}{y-x+2}$

$$1 - z' = \frac{-x-1-z-x+x}{x-1-z-x+2}$$

$$1 - z' = \frac{-z}{-z+1} = \frac{z}{z-1}$$

$$1 - \frac{z}{z-1} = z' = \frac{dz}{dx}$$

$$\Downarrow$$

$$\left| 1 - \frac{z}{z-1} \right. \qquad (z-1)\,z' = 1$$

$$\frac{z-1-z}{z-1} = z' \Rightarrow \frac{-1}{z-1} = \frac{dz}{dx} \Rightarrow$$

$$\Rightarrow (z-1)\,dz = -dx \overset{\int}{\Rightarrow} \int (z-1)\,dz = -\int dx$$

$$\Rightarrow \frac{z^2}{2} - z = -x + c \mid 2$$

$$\Rightarrow z^2 - 2z = -2x + 2c$$

$$\Rightarrow (x-y-1)^2 - 2(x-y-1)^2 = -2x + 2c$$

$$\Leftrightarrow \boxed{x^2 + y^2 - 2xy + 4y - 6x = k}$$

(ex) SEMINAR

$(2x+y+1)dx - (4x+2y-3)dy = 0$

$y' = \frac{dy}{dx}$

$(2x+y+1)dx = (4x+2y-3)dy$

$$\Downarrow$$

$$y' = \frac{2x+y+1}{4x+2y-3} = \frac{2x+y+1}{2(2x+y+1)-5}$$

Dacă $z(x) = 2x + y(x) + 1 \Rightarrow z' = 2 + y' \Rightarrow y' = z' - 2$

Înlocuiesc: $z' - 2 = \frac{z}{2z-5}$

$$z' = \frac{z}{2z-5} + 2 = \frac{z+4z-10}{2z-5} = \frac{5z-10}{2z-5}$$

$z' = \frac{5(z-2)}{2z-5}$ ec cu var separabile

$\frac{dz}{dx} = \frac{5(z-2)}{2z-5} \Rightarrow \frac{2z-5}{(z-2)}\,dz = 5\,dx$

$\int$

$\Rightarrow \int \frac{2z-5}{(z-2)}\,dz = 5\int dx$

$\Rightarrow \int \left(2 - \frac{1}{z-2}\right) dz = 5x + c$

$\Rightarrow 2z - \ln|z-2| = 5x + c$

$\Rightarrow 2(2x+y(x)+1) - \ln|2x+y(x)-1| = 5x + c$

$\Rightarrow 2y + 2 - x = c + \ln|2x+y-1|$

$\Rightarrow 2y + 2 - x = \ln e^c + \ln(2x+y-1)$

$\Rightarrow \ldots$

10.2.2022

EX TIP 1 SEMINAR

1) Să se dea soluția generală

a) $x\,dy - y\,dx = \sqrt{xy}\,dx$, $x, y > 0$

$y' = \frac{dy}{dx}$

$y' = f\left(\frac{y}{x}\right)$

$\Rightarrow x \cdot y' - y = \sqrt{xy}$ | $:x$

$\Rightarrow y' = \frac{\sqrt{xy} + y}{x}$

$\Rightarrow y' = \frac{\sqrt{xy}}{x^2} + \frac{y}{x} = \sqrt{\frac{y}{x}} + \frac{y}{x}$

Se mai scrie $y'(x) = f\left(\frac{y(x)}{x}\right)$, unde $f(t) = t + \sqrt{t}$

• Fac schimbarea de var.

$z(x) = \frac{y(x)}{x} \Rightarrow y(x) = z(x) \cdot x \Rightarrow$

$\Rightarrow y'(x) = z'(x)(x) + x' \cdot z(x)$

$\Rightarrow y'(x) = x \cdot z' + z$

• Înlocuire

$x \cdot z' + z = \sqrt{z} + z$ (EVS)

$x \frac{dz}{dx} = \sqrt{z} + z - z$

$x \frac{dz}{dx} = \sqrt{z} \Rightarrow \frac{dz}{\sqrt{z}} = \frac{1}{x}dx$

$\int \Rightarrow \int \frac{1}{\sqrt{z}}\,dz = \int \frac{1}{x}\,dx$

$\Rightarrow 2\sqrt{z} = \ln x + c$

$\Rightarrow z = \left(\frac{\ln x + c}{2}\right)^2$

$\Rightarrow \frac{y}{x} = \left(\frac{\ln x + c}{2}\right)^2 \Rightarrow$

$\Rightarrow y = x\left(\frac{\ln x + c}{2}\right)^2$.

"TIP 3"

$y' = f\left(\frac{a_1x + b_1y + c_1}{a_2x + b_2y + c_2}\right)$, $f \in C^1$.

$\Delta \overset{not}{=} \begin{vmatrix} a_1 & b_1 \\ a_2 & b_2 \end{vmatrix} \neq 0 \Rightarrow \exists!\,(x_0, y_0)$ sol. sist $\begin{cases} a_1x + b_1y + c_1 = 0 \\ a_2x + b_2y + c_2 = 0 \end{cases}$

Schimbarea de funcție: $z(x) = \frac{y(x) - y_0}{x - x_0}$

EX CURS

$y'(x+y-1)^2 = 2(y+2)^2$

$\Downarrow$

$y' = \frac{2(y+2)^2}{(x+y-1)^2} \Rightarrow \Delta = \begin{vmatrix} 0 & 1 \\ 1 & 1 \end{vmatrix} = -1 \neq 0$

$\begin{cases} y + 2 = 0 \\ x + y = 1 \end{cases} \Rightarrow \begin{cases} y = -2 \\ x = 3 \end{cases}$ $(x_0, y_0) = (3, -2)$

$z(x) = \frac{y(x) + 2}{x - 3} \Leftrightarrow y(x) = z(x-3) - 2$

$\Rightarrow y'(x) = z'(x-3) + z(x-3)'$

$y'(x) = z'(x-3) + z$

Înlocuire

$z'(x-3) + z = 2\,\frac{z^2(x-3)^2}{(x + z(x-3) - 3)^2}$

$[(z+1)(x-3)]^2$

$z'(x-3) + z = 2\,\frac{z^2}{(z+1)^2}$

$z'(x-3) = 2\,\frac{z^2}{(z+1)^2} - z$ (EVS)

$\frac{dz}{dx}(x-3) = \frac{2z^2}{(z+1)^2} - z$ | $\cdot \frac{1}{dz}$

$\frac{dz}{dx}(x-3) = \frac{z - z^3}{(z+1)^2} \Rightarrow \frac{x-3}{dx} = \frac{z - z^3}{(z+1)^2}\,dz$

ex SEMINAR

$(2x-4y+6)\,dx + (x+y-3)\,dy = 0.$

$y' = \frac{dx}{dy} = \frac{2x-4y+6}{x+y-3}$

$\Delta = \begin{vmatrix} 2 & -4 \\ 1 & 1 \end{vmatrix} = 2+4 = 6 \neq 0$

$\begin{cases} x-2y+3=0 \\ x+y-3=0 \mid \cdot 2 \end{cases} \Rightarrow \begin{cases} x-2y+3=0 \\ 2x+2y-6=0 \end{cases}$ (+)

$3x-3=0 \Rightarrow x=1 \Rightarrow y=2.$

$(x_0, y_0) = (1,2)$

$z(x) = \frac{y-2}{x-1} \Rightarrow y(x) = z(x-1)+2$

$\Rightarrow y'(x) = z'(x-1) + z$

(*) $\begin{cases} z'(x-1)+z = \frac{2x-4(z(x-1)+2)+6}{x+z(x-1)-1} \\ z'(x-1)+z = \frac{2x-4z(x-1)-2}{x+z(x-1)-1}. \end{cases}$

Fac un mic „artificiu"

crad că s-a greșit la calcul

$y' = -\frac{2x-4y+6}{x+y-3} = \frac{-2(x+1)-4(y-2)}{x-1+y-2} = -\frac{2-4\frac{y-2}{x-1}}{1+\frac{y-2}{x-1}}$

Continuarea în următoarea și ajungi la EVS

III) Ecuații diferențiale liniare de ordinul I

Forma $y' + f(x)\cdot y = g(x)$

• înmulțesc cu $e^{\int f(x)dx}$

ex CURS 1) $x(x-1)y' + 2xy = 1$, $x > 1$

$\left(y' = \frac{1-2xy}{x(x-1)} = \right)$

$y' + \underbrace{\frac{2}{x-1}}_{f(x)} y = \underbrace{\frac{1}{x(x-1)}}_{g(x)} \mid \cdot e^{\int \frac{2}{x-1}dx} = e^{2\ln(x-1)} = (x-1)^2$

$\Downarrow$

$y'(x-1)^2 + y\cdot\frac{2}{x-1}(x-1)^2 = \frac{(x-1)^2}{x(x-1)}$

$\Downarrow$

$y'(x-1)^2 + 2(x-1)y = \frac{x-1}{x}$

• observ că obțin $\left(y\cdot e^{\int f(x)dx}\right)'$

$\left(y\cdot(x-1)^2\right)' = \frac{x-1}{x}$

$\Downarrow S$

• integrez $y(x-1)^2 = \int \frac{x-1}{x}dx$

$(x-1)^2 y = \int \left(1-\frac{1}{x}\right)dx$

$\Downarrow$

$(x-1)^2 y = x - \ln x + C$, $C \in \mathbb{R}$

$\Downarrow$

$y = \frac{1}{(x-1)^2}(x-\ln x + C)$, $C \in \mathbb{R}$

SEMINAR

2) $(xy+e^x)dx - xdy = 0$

$\Downarrow$

$y' = \frac{dy}{dx} = \frac{xy+e^x}{x}$

$\Downarrow$

$y' x - xy = e^x \mid :x$

$\Downarrow$

$y' - y = \frac{e^x}{x}$

$\Downarrow \quad \cdot e^{-\int dx} = e^{-x}$

$y' e^{-x} - y e^{-x} = \frac{1}{x}$

$\Downarrow$

$(y \cdot e^{-x})' = \frac{1}{x}$

$\Downarrow \int$

$y e^{-x} = \ln|x| + c$

$\Downarrow$

$y = (\ln|x| + c) e^x$

3) YOUTUBE

$y' + xy = x e^{-\frac{x^2}{2}}$

$\Downarrow \quad \cdot e^{\int x\,dx}$

$e^{\frac{x^2}{2}} y' + x e^{\frac{x^2}{2}} \cdot y = x e^{-\frac{x^2}{2}} \cdot e^{\frac{x^2}{2}}$

$\Downarrow$

$\left(y \cdot e^{\frac{x^2}{2}}\right)' = x$

$\Downarrow \int$

$y \cdot e^{\frac{x^2}{2}} = \frac{x^2}{2} + c \quad \Rightarrow \quad y = \left(\frac{x^2}{2} + c\right) e^{-\frac{x^2}{2}}$

4) $y' + y \sin x = -\sin x \cos x \qquad (\sin x)' = \cos x$

$\Downarrow \quad \cdot e^{\int \sin x\,dx} = e^{-\cos x}$

$y' e^{-\cos x} + y \sin x\, e^{-\cos x} = -e^{-\cos x} \sin x \cos x$

$\Downarrow$

$(y \cdot e^{-\cos x})' = -e^{-\cos x} \sin x \cos x$

$\Downarrow \int$

$y \cdot e^{-\cos x} = -\int e^{-\cos x} \sin x \cos x\,dx$

$\Downarrow \qquad (\sin x)' = \cos x$
$(\cos x)' = -\sin x$

$y e^{-\cos x} = \int e^{-\cos x} (\cos x)' \cos x\,dx \overset{\cos x = t}{=}$

$\int e^{-t} t\,dt = -\int (e^{-t})' \cdot t\,dt = -e^{-t} t - e^{-t} = -e^{-t}(t+1)$

$= -e^{-\cos x} \cos x - e^{-\cos x} + c$

$y = -\cos x - 1 + c e^{\cos x}$

IV) Ecuații diferențiale BERNOULLI

Forma generală: $\boxed{y' + f(x) \cdot y = g(x) y^\alpha}$ $\mid \cdot y^{-\alpha}$

$y' y^{-\alpha} + f(x) \underbrace{y^{1-\alpha}}_{z} = g(x)$

- se reduce la o ecuație liniară

Fac schimbarea $z(x) = y^{1-\alpha}(x)$

CURS

EX 1) $y' + xy = \frac{e^{-2x^2}}{y^3}$

$\Downarrow$

$y' + xy = y^{-3} e^{-2x^2} \mid \cdot y^3$ • Înmulțesc (dacă e nev. ai $y^{-\alpha}$)

$y' y^3 + x \underbrace{y^4}_{z} = e^{-2x^2}$ • $z(x) = y^{1-\alpha}$ $\alpha$

Fac schimbarea de var. $z(x) = y^{1-\alpha} \Rightarrow z' = (1-\alpha) y^{-\alpha} \cdot y'$

$z(x) = y^4 \Rightarrow z' = 4y^3 y'$

$x^3 y' + x y^4 = e^{-2x^2} \quad | \cdot 4$ · înmulțesc cu coef. afferent pt a îmi da $z'$

$\Downarrow$

$4y^3 y' + 4x y^4 = 4e^{-2x^2}$

$\Downarrow$

$z' + \underbrace{4}_{f(x)} z = \underbrace{4e^{-2x^2}}_{g(x)}$ · ajung la o ecuație liniară de ordinul I

$\Downarrow \cdot e^{\int 4x dx} = e^{2x^2}$

$z' e^{2x^2} + 4z x e^{2x^2} = 4$

$\Downarrow$

$(z e^{2x^2})' = 4$

$\Downarrow \int$

$z e^{2x^2} = 4x + C$

$\Downarrow$

$z = (4x+C) e^{-2x^2}$

$\Downarrow$

$y^4 = (4x+C) e^{-2x^2}$

· SEMINAR

2) $2y' - \frac{x}{x^2-1} y = \frac{x}{x^2-1}$

$\Downarrow$

$y' - \frac{1}{2} \frac{x}{x^2-1} y = \frac{x}{x^2-1} y \frac{1}{2}$

$y' - \frac{1}{2} \frac{x}{x^2-1} y = \frac{1}{2} \frac{x}{x^2-1} \cdot y^{1-\alpha}$

$\Rightarrow y' y^{-\alpha} - \frac{1}{2} \frac{x}{x^2-1} y^{1-\alpha} = \frac{1}{2} x \quad | \cdot 2$

$z = y^{1-\alpha} \Rightarrow z'(x) = (1-\alpha) y^{-\alpha} y'$

$\Rightarrow 2 y' y^{-\alpha} - \frac{x}{x^2-1} y^{1-\alpha} = x$

$\Downarrow$

$z' - \frac{x}{x^2-1} z = x$ ecuație liniară de ord I

$\Downarrow e^{-\int \frac{x}{x^2-1} dx} = e^{-\ln \sqrt{x^2-1}} = \frac{1}{\sqrt{x^2-1}}$

$z' \frac{1}{\sqrt{x^2-1}} - \frac{x}{(x^2-1)^{3/2}} z = \frac{x}{\sqrt{x^2-1}}$

$\Downarrow$

$\left( z \frac{1}{\sqrt{x^2-1}} \right)' = \frac{x}{\sqrt{x^2-1}}$

$\Downarrow \int$

$z \frac{1}{\sqrt{x^2-1}} = \int \frac{x}{\sqrt{x^2-1}} dx$

$\Downarrow$

$\frac{z}{\sqrt{x^2-1}} = \sqrt{x^2-1} + c$

$\Downarrow$

$z = x^2 - 1 + c\sqrt{x^2-1}$

$\Downarrow$

$y^2 = x^2 - 1 + c\sqrt{x^2-1}$

3) $y' = y^4 \cos x + y \operatorname{tg} x, \quad x \in (0, \frac{\pi}{2})$

$y' - y \operatorname{tg} x = y^4 \cos x \quad | \cdot y^{-4}$

$y' \cdot y^{-4} - y^{-3} \operatorname{tg} x = \cos x \quad | \cdot (-3)$

$z(x) = y^{-3}(x) \Rightarrow z'(x) = (-3) y^{-4} y'$

$-3y' y^{-4} + 3y^{-3} \operatorname{tg} x = -3\cos x$

$\Downarrow$

$z' + 3z \operatorname{tg} x = -3\cos x$ ec liniară

$\Downarrow$ $e^{3\int \operatorname{tg} x\,dx} = e^{-3\ln(\cos x)} = \frac{1}{(\cos x)^3}$

$z' \frac{1}{(\cos x)^3} + 3\frac{\sin x}{(\cos x)^4} z = -3\frac{1}{(\cos x)^3}$

$\Downarrow$

$\left(z \frac{1}{\cos^3 x}\right)' = \frac{-3}{\cos^3 x}$

$\Downarrow \int$

$z \frac{1}{\cos^3 x} = -3\int \frac{1}{\cos^3 x} dx$

$\Downarrow$

$z \frac{1}{\cos^3 x} = -3\operatorname{tg} x + c$

$\Downarrow$

$z = -3\operatorname{tg} x \cos^3 x + c\cos^3 x$

$\Downarrow$

$y^{-3} = -3\operatorname{tg} x \cos^3 x + c\cos^3 x$.

V) Ecuații dif. liniare omogene de ordin superior cu coef. ct.

- rezolv folosind ecuația caracteristică

i) $\boxed{\lambda = \alpha \in \mathbb{R}}$ cu $m_a \geq 1$ → am m soluții L I

$e^{rx}, xe^{rx}, x^2e^{rx}, \ldots, x^{m-1}e^{rx}$

ii) $\boxed{\lambda = \alpha \pm \beta i}$ cu $s \geq 1$ → am 2s soluții L I

mult alg.

$e^{\alpha x}\cos\beta x, \; x e^{\alpha x}\cos\beta x, \; x^2 e^{\alpha x}\cos\beta x, \ldots, x^{s-1} e^{\alpha x}\cos\beta x$

$e^{\alpha x}\sin\beta x, \; x e^{\alpha x}\sin\beta x, \; x^2 e^{\alpha x}\sin\beta x, \ldots, x^{s-1} e^{\alpha x}\sin\beta x$

Formula Euler $e^{(\alpha+i\beta)x} = e^{\alpha x} e^{i\beta x} = e^{\alpha x}(\cos\beta x + i\sin\beta x)$

. Soluția generală

$y(x) = c_1 y_1(x) + c_2 y_2(x) + \ldots + c_n y_n(x)$

ex curs

a) $y^{(5)} - y^{(4)} + y^{(3)} = 0$

. Rezolv ec caract:

$\lambda^5 - \lambda^4 + \lambda^3 = 0 \Rightarrow \lambda^3(\lambda^2 - \lambda + 1) = 0$

$\Rightarrow \begin{cases} \lambda_1 = 0, \; m = 3 \\ \lambda_{2,3} = \frac{1}{2} \pm i\frac{\sqrt{3}}{2} \end{cases}$ cu $\alpha = \frac{1}{2}$, $\beta = \frac{\sqrt{3}}{2}$, $n_1 = n_2 = 1$

$\boxed{\lambda = 0} \Rightarrow e^{0x}, xe^{0x}, x^2 e^{0x} \Rightarrow 1, x, x^2$

$\boxed{\lambda_{2,3} = \frac{1}{2} \pm i\frac{\sqrt{3}}{2}} \Rightarrow e^{\frac{1}{2}x}\cos\left(\frac{\sqrt{3}}{2}x\right), \; e^{\frac{1}{2}x}\sin\left(\frac{\sqrt{3}}{2}x\right)$

Deci $y(x) = c_1 \cdot 1 + c_2 \cdot x + c_3 x^2 + c_4 e^{\frac{1}{2}x}\cos\frac{\sqrt{3}}{2}x + c_5 e^{\frac{1}{2}x}\sin\frac{\sqrt{3}}{2}x$ cu ct reale.

TREC

VI) Ecuații dif. liniare necomogene de ordin superior cu coef. constanți

$y(x) = y_h(x) + y_p(x)$

## Determinarea lui $y_p$

$\cos 0 = 1$

$$f(x) = e^{ax}\left(P_m(x)\cos bx + Q_m(x)\sin bx\right)$$

$N = \max\{m_1, m_2\}$ dacă $\alpha \in \mathbb{N}$ mult. alg. a lui $\boxed{0+\beta i}$ ca rădăcină a ec. caracteristice ($S=0$, dacă nu e rădăcină)

Ideea și restul

$$\boxed{y_p = x^S e^{ax}\left(\widetilde{P}_N(x)\cos bx + \widetilde{Q}_N(x)\sin bx\right)}$$

polinoame de grad $\le N$

CURS

Ex 1) $y'' - 9y = e^{3x}\cos x \quad (1)$

pas 1: Rezolv ec. caract. $f(x)$

$\lambda^2 - 9 = 0 \Leftrightarrow \lambda_{1,2} = \pm 3$ cu $m_{1,2} = 1$

$\Rightarrow$ Sol fundamentală $e^{3x}, e^{-3x}$

$\Rightarrow \bullet\ y_h = c_1 e^{3x} + c_2 e^{-3x}$ cu $c_1, c_2 \in \mathbb{R}$

2) Det $y_p$. $f(x) = e^{3x}\cos x$

$\left.\begin{matrix} a = 3 \\ b = 1 \end{matrix}\right\} \Rightarrow a + bi = 3 + i$ nu e rădăcină a ecuației caract

$\Rightarrow \boxed{S=0}$.

$P_m(x) = 1 \Rightarrow m = 0$

$Q_m(x) \overset{\text{alg}}{=} 0 \Rightarrow m = -\infty$ $\Big| \Rightarrow N = 0$.

$y_p = x^S e^{ax}\left(\widetilde{P}_N(x)\cos bx + \widetilde{Q}_m(x)\sin bx\right)$

$= x^0 e^{3x}\left(\widetilde{P}_0(x)\cos x + \widetilde{Q}_0(x)\sin x\right)$

$\Rightarrow y_p = \underbrace{e^{3x}}_{f}\left(\underbrace{\alpha\cos x + \beta\sin x}_{g}\right)$ $\quad (fg)'' = f''g + 2f'g' + fg''$

3) Determin coef în ecuația dif. dată

$y_p'' = ge^{3x}(\alpha\cos x + \beta\sin x) + 6e^{3x}(-\alpha\sin x + \beta\cos x) + e^{3x}(-\alpha\cos x - \beta\sin x)$

Înlocuiesc în (1) obțin că

$ge^{3x}(-\alpha\sin x + \beta\cos x) + e^{3x}(-\alpha\cos x - \beta\sin x) = e^{3x}\cos x$

$\Downarrow$ $\{\sin x, \cos x\}$ SLI

mă uit după sin și cos

$$\begin{cases} -6\alpha - \beta = 0 \\ 6\beta - \alpha = 1 \end{cases} \Leftrightarrow \begin{cases} \beta = -6\alpha \\ \alpha = -\frac{1}{37} \end{cases} \Rightarrow \beta = \frac{6}{37}$$

Deci $y_p = \frac{e^{3x}}{37}\left(-\cos x + 6\sin x\right)$

Deci $\boxed{y = y_p + y_h}$

Ex 2 CURS

$y'' - 5y' = \underbrace{3x^2}_{y_{p_1}} + \underbrace{\sin 5x}_{y_{p_2}}$

Sumă de cvasipolinoame

pas 1 $\quad \lambda^2 - 5\lambda = 0 \Rightarrow \lambda(\lambda - 5) = 0 \Rightarrow \begin{cases} \lambda_1 = 0, & m_1 = 1 \\ \lambda_2 = 5, & m_2 = 1 \end{cases}$

$\Rightarrow e^{0x} = 1$ ; $e^{5x}$ soluții fundamentale

$\Rightarrow y_h = c_1 \cdot 1 + c_2 e^{5x}$, $c_1, c_2 \in \mathbb{R}$

• $\boxed{y_{p_1} =}$ $y'' - 5y' = \underbrace{3x^2}_{f(x)}$ (1)

$a = 0$

$b = 0$ $\Rightarrow a + bi = 0$ soluție a ec. caracteristice cu $\boxed{\alpha = 1}$

$P_m(x) = 3x^2 \Rightarrow m = 2$

$Q_m(x) \overset{\text{alg}}{=} 0 \Rightarrow m = -\infty$ $\Big| \Rightarrow N = 2$.

Caso $y_p^1 = x^S e^{ax}(\tilde{P}_M(x)\cos bx + \tilde{Q}_N(x)\sin bx)$

$= x\,(\tilde{P}_2(x)\cos(0x) + \tilde{Q}_2(x)\sin(0x))$

$= x(cx^2+dx+e) = cx^3+dx^2+ex$

$(y_p^1)' = 3cx^2 + 2dx + e$

$(y_p^1)'' = 6cx + 2d$

Substituir em (A)

$6cx + 2d - 15cx^2 - 10dx + 5e = 3x^2$

$\Rightarrow \begin{cases} c = -\frac{1}{5} \\ 6c - 10d = 0 \Rightarrow d = -0{,}12 \\ e = \frac{2}{5}d \Rightarrow e = -0{,}048 \end{cases}$

Logo $y_p^1 = -\frac{1}{5}x^3 - 0{,}12x^2 - 0{,}048x$

- $y_p^2$ $\quad y'' - 5y' = \sin 5x = e^{ax}(P_m(x)\cos bx + Q_m(x)\sin bx)$

$\begin{cases} a = 0 \\ b = 5 \end{cases} \Rightarrow a + ib = 5i$ não é raiz da eq caract $\Rightarrow S = 0$

$P_m(x) \overset{\text{alg}}{=} 0 \Rightarrow m = -\infty$

$Q_m(x) = 1 \Rightarrow m = 0$ $\Big\} \Rightarrow N = 0$

$y_p^2(x) = x^0 \cdot e^{0x}(\tilde{P}_0(x)\cos 5x + \tilde{Q}_0(x)\sin 5x) =$

$= g\cos 5x + h\sin 5x$

$(y_p^2)' = 5g - \sin 5x + 5h\cos 5x$

$(y_p^2)'' = -25g\cos 5x - 25h\sin 5x$

Substituir $\Rightarrow$

$\Rightarrow -25g\cos 5x - 25h\sin 5x + 25g\sin 5x - 25h\cos 5x = \sin 5x$

$\Downarrow$ $h\cos x, \sin x$ SLI

$\Rightarrow \begin{cases} g + h = 0 \quad | \cdot 25 \\ -25h + 25g = 1 \end{cases} \Rightarrow \begin{cases} g = \frac{1}{50} \\ h = -\frac{1}{50} \end{cases}$

Logo $y_p^2 = \frac{1}{50}\cos 5x - \frac{1}{50}\sin 5x$

Logo $\boxed{y = y_h + y_p^1 + y_p^2}$

Ex 2 Sabe-se que a sol geral da ec dif

$y'' + 2y' + y = xe^x + \frac{1}{xe^x}$

$\lambda^2 + 2\lambda + 1 = 0$

$\Updownarrow$

$(\lambda+1)^2 = 0 \iff \lambda = -1$ ou $m = 2$

Solução fundamental: $e^{-x}$, $xe^{-x}$

$y_h(x) = c_1 e^{-x} + c_2 x e^{-x} = e^{-x}(c_1 + xc_2)$, $c_1, c_2 \in \mathbb{R}$

$y_p^1$ $\quad y'' + 2y' + y = \underbrace{xe^x}_{f(x)}$

$f(x) = e^{ax}(P_m\cos bx + Q_m\sin bx)$

$\begin{cases} a = 1 \\ b = 0 \end{cases} \Rightarrow a + bi = 1$ NÃO é sol da eq caract

$\Rightarrow S = 0$

$P_m(x) = x$
$Q_m(x) \overset{\text{alg}}{=} 0$ $\Big\}$ $\Rightarrow$ $\boxed{N=1}$

$y_p^1 = x^s e^{ax} (\tilde{P}_N(x)\cos bx + \tilde{Q}_N(x)\sin bx)$

$y_p^1 = e^x(cx+d)$

$(y_p^1)' = e^x(cx+d+c)$

$(y_p^1)'' = e^x(cx+d+2c)$

Înlocuesc

$\Rightarrow e^x(cx+d+2c+2cx+2d+2c+cx+d) = x e^x$

$\Rightarrow 4cx + 4d + 4c = x$

$\Rightarrow 4c = 1 \Rightarrow c = \frac{1}{4} \Rightarrow d = -\frac{1}{4}$.

Deci $y_p^1 = \frac{e^x}{4}(x-1)$

$y_p^2$ $y'' + 2y' + y = \frac{1}{xe^x}$ $f(x)$

Folosesc metoda variației constantelor

· transformăm constantele în funcții

Caut $y_{p_2}$ de forma:

$y_p^2 = c_1(x)y_1(x) + c_2(x)y_2(x)$ → pt că are grad 2 ca $y$

$y_1 = e^{-x}$, $y_2 = xe^{-x}$

$\begin{cases} c_1' y_1 + c_2' y_2 = 0 \\ c_1' y_1' + c_2' y_2' = \frac{1}{x} e^{-x} \end{cases}$ $\frac{f(x)}{a_n}$, an este coef lui $y''$

$\Leftrightarrow \begin{cases} c_1' e^{-x} + c_2' x e^{-x} = 0 \\ -c_1' e^{-x} + c_2'(e^{-x} - xe^{-x}) = \frac{1}{x} e^{-x} \end{cases}$

$e^{-x}(1-x)$

$\Leftrightarrow \begin{cases} c_1' + c_2' x = 0 \\ -c_1' + c_2'(1-x) = \frac{1}{x} \end{cases}$

$c_2' = \frac{1}{x} \overset{\int}{\Rightarrow} c_2(x) = \ln|x| + A_2$ ① $\Big\}$ $A_1 = A_2 = 0$ $\Rightarrow$

$\Rightarrow c_1' = -1 \overset{\int}{\Rightarrow} c_1(x) = -x + A_1$

$\Rightarrow y_p^2(x) = \cancel{\ln|x| \cdot e^{-x} \cdot (x) \cdot e^{-x}} - x e^{-x} + (\ln|x|) \cdot xe^{-x}$

Deci $\boxed{y = y_n + y_{p_1} + y_{p_2}}$

## VII) Ecuații de EULER

· $x$ ridicat la o pt liniară cu coef $\alpha$

· recunoaștere după gradele egale (ex $x^2y'' - xy' + y$)

ex $x^2y'' - 2y = \frac{3x^2}{x+1}$

1) Fac substituția schimbării

$x = e^t$ și $z(t) = y(e^t)$

Am atunci $\begin{cases} y'(x) = e^{-t} \cdot z'(t) \\ y''(x) = e^{-2t}(z''(t) - z'(t)) \end{cases}$

2) Înlocuiesc $e^{2t} e^{-2t}(z'' - z') - 2z = \frac{3e^{t2}}{e^t + 1}$

$\Rightarrow z'' - z' - 2z = \frac{3e^{2t}}{1+e^t} = f(t)$ ec. dif. lineară cu coef. ct., rezolvată în cadrul temei II

$\lambda^2 - \lambda - 2 = 0$

$\Delta = 9 \Rightarrow \lambda_{1,2} = \frac{1 \pm 3}{2}$ $\begin{cases} \lambda_1 = 2 \Rightarrow e^{2t} = z_1(t) \\ m_1 = 1 \\ \lambda_2 = -1 \Rightarrow e^{-t} = z_2(t) \\ m_2 = 1 \end{cases}$

$\Rightarrow z_h(t) = c_1 e^{2t} + c_2 e^{-t}$ cu $c_1, c_2 \in \mathbb{R}$

3) Observăm că nu am rezolvat pe larg, deci fol. met. var. lui t

Fie $z_p(t) = c_1(t) \underbrace{e^{2t}}_{z_1} + c_2(t) \underbrace{e^{-t}}_{z_2}$, unde $c_1'$ și $c_2'$ verif.

$$\begin{cases} c_1' z_1 + c_2' z_2 = 0 \\ c_1' z_1' + c_2' z_2' = \frac{3e^{2t}}{1+e^t} \end{cases}$$

$$\Downarrow$$

$$\begin{cases} c_1' e^{2t} + c_2' e^{-t} = 0 \\ 2c_1' e^{2t} - c_2' e^{-t} = \frac{3e^{2t}}{1+e^t} \end{cases}$$

$\oplus$

$$3c_1' e^{2t} = \frac{3e^{2t}}{1+e^t}$$

$$\Downarrow$$

$$c_1(x) = \int \frac{1}{1+e^t}\,dt \quad \begin{matrix} e^t = u \\ \hline du = e^t dt \end{matrix} \quad \int \frac{1}{u+1} \cdot \frac{1}{u}\,du$$

$$= \int \frac{u+1-u}{u(u+1)}\,du = \int \frac{1}{u} - \frac{1}{u+1}\,du = \ln\left|\frac{u}{u+1}\right| + A_1 =$$

$$= \ln\left|\frac{e^t}{e^t+1}\right| + A_1 = t - \ln(1+e^t) + A_1$$

$$c_2'(t) = -e^{3t} \quad c_1'(t) = -\frac{e^{3t}}{1+e^t}$$

$$\Downarrow$$

$$(*) \begin{cases} c_2(t) = -\int \frac{e^{3t}}{1+e^t}\,dt \end{cases}$$

$$c_2' e^{-t} = -c_1' e^{2t}$$

$$\Downarrow$$

$$c_2' e^{-t} = -\frac{1}{1+e^t} e^{2t} \quad \Big| \cdot \frac{1}{e^{-t}}$$

$$\Downarrow$$

$$c_2' = -\frac{e^{2t}}{1+e^t} \cdot e^t = -\frac{e^{3t}}{1+e^t}$$

$$\Downarrow \int$$

$$c_2 = -\int \frac{e^{3t}}{1+e^t}\,dt =$$

În final, $z = z_h + z_p$, iar apoi înlocuiesc în $y(x)$

★ LAST CHAPTER
KEEP GOING

TRICI

# Sisteme

1) Sisteme cu coef. constanți

- A diagonalizabilă $\Rightarrow$ sist diff $X'(t) = A\,X(t)$

a) $\lambda \in \mathbb{R}$ cu mult alg $\geq 1$ are loc $\dim_{\mathbb{R}} V_\lambda = m$, $\lambda \in S(A)$ $\Rightarrow \{v_1, \dots, v_m\}$ is. liniar

Avem $m$ soluții L I

$e^{\lambda t} v_1, e^{\lambda t} v_2, \dots, e^{\lambda t} v_m$

b) $\alpha \pm i\beta \in S(A)$ cu mult alg $s$ are loc $\dim_{\mathbb{R}} V_{\alpha+\beta} = s$

Avem $2s$ soluții L I — $\Rightarrow \{w_1, w_2, \dots, w_n\}$ bază

$\mathrm{Re}(e^{(\alpha+i\beta)t} w_1), \dots, \mathrm{Re}(e^{(\alpha+i\beta)t} w_m)$

$\mathrm{Im}(e^{(\alpha+i\beta)t} w_1), \dots, \mathrm{Im}(e^{(\alpha+i\beta)t} w_s)$

Obs Dem Euler $e^{it} = \cos t + i \sin t$

EX (CURS)

$$\begin{cases} x_1' = 2x_1 + x_2 \\ x_2' = x_1 + 3x_2 - x_3 \\ x_3' = -x_1 + 2x_2 + 3x_3 \end{cases} \Rightarrow X'(t) = A\,X(t)$$

1) aflu vectorii și valorile proprii

$$A = \begin{pmatrix} 2 & 1 & 0 \\ 1 & 3 & -1 \\ -1 & 2 & 3 \end{pmatrix}$$

$$P_A(\lambda) = \det(A - \lambda I_3) = \begin{vmatrix} 2-\lambda & 1 & 0 \\ 1 & 3-\lambda & -1 \\ -1 & 2 & 3-\lambda \end{vmatrix} \quad C_1 \to C_1 + C_3$$

$$= \begin{vmatrix} 2-\lambda & 1 & 0 \\ 0 & 3-\lambda & -1 \\ 2-\lambda & 2 & 3-\lambda \end{vmatrix} = (2-\lambda) \begin{vmatrix} 1 & 1 & 0 \\ 0 & 3-\lambda & -1 \\ 1 & 2 & 3-\lambda \end{vmatrix} \quad C_{(2)} \to C_2 - C_1$$

$$= (2-\lambda) \begin{vmatrix} 1 & 0 & 0 \\ 0 & 3-\lambda & -1 \\ 1 & 1 & 3-\lambda \end{vmatrix} = (2-\lambda)\left((3-\lambda)^2 + 1\right)$$

$\lambda_1 = 2$, cu $m_{\lambda_1} = 1$

$(3-\lambda)^2 + 1 = 9 - 6\lambda + \lambda^2 + 1 = \lambda^2 - 6\lambda + 10$

$\lambda_{2,3} = 36 - 40 = -4$

$\lambda_{2,3} = \frac{6 \pm 2i}{2}$ $\Rightarrow$ $\lambda_1 = 3+i$, $\lambda_2 = 3-i$ cu $S_{2,3} = 1$

$\Rightarrow$ A admite valori proprii simple $\Rightarrow$ A diagonalizabilă peste $\mathbb{C}$

$\lambda_1 = 2$ Caut $v \in (a,b,c)$ vector propriu asociat

$Av^t = \lambda_1 v^t$

$$\begin{pmatrix} 2 & 1 & 0 \\ 1 & 3 & -1 \\ -1 & 2 & 3 \end{pmatrix} \begin{pmatrix} a \\ b \\ c \end{pmatrix} = \begin{pmatrix} 2a \\ 2b \\ 2c \end{pmatrix} \Rightarrow \begin{cases} 2a + b = 2a \Rightarrow b = 0 \\ a + 3b - c = 2b \Rightarrow c - a = b \\ -a + 2b + 3c = 2c \end{cases}$$

$\Downarrow$ $a = c$

$c - a = b$

$-c - a = 2b$

$-2a = 3b \Rightarrow a = -\frac{3}{2}b$

Fie $a = \alpha \in \mathbb{R} \Rightarrow v \in \{(\alpha, 0, \alpha)\} = \{\alpha(1,0,1)\} = V_{\lambda_1}$ — $v_1$ subs. propriu asociat

$\Rightarrow \{v_1\}$ sist. o bază pt un vect. subs. $V_{\lambda=2}$.

$\lambda_2 = 3+i$ Caut $v \in (a,b,c)$ vector propriu asociat

$$\begin{cases} 2a + b = (3+i)a \\ a + 3b - c = (3+i)b \\ -a + 2b + 3c = (3+i)c \end{cases} \Rightarrow \begin{cases} 2a + b - 3a - ai = 0 \\ a + 3b - c = 3b + bi \\ -a + 2b + 3c = 3c + ci \end{cases}$$

$$\Rightarrow \begin{cases} -a + b - ai = 0 \\ a - c - bi = 0 \\ -a + 2b - ci = 0 \end{cases}$$

$\begin{cases} a-c-b\,i=0 \\ -a+b-a\,i=0 \\ -a+2b-c\,i=0 \end{cases}$ ① ② ③

② $\quad b-c-i(a+b)=0 \Rightarrow$

③ $\quad 2b-c-i(b+c)=0$

$-(i+1)(1+i) = -(i^2+2i+1)$

$= 1+2i-1$

Nova base $\mathcal{C}$

Encontrar a matriz

$A-\lambda_2 I_3 = \begin{pmatrix} -1-i & -1 & -1 \\ 1 & -1 & -1-i \\ -1 & 2 & -i \end{pmatrix} \longrightarrow \begin{pmatrix} 1 & -1 & -1 \\ -1-i & 1 & 0 \\ -1 & 2 & -i \end{pmatrix} \begin{matrix} \\ (i+1) \\ \\ \end{matrix}$ ① ②

$\begin{pmatrix} 1 & -1 & -1 \\ 0 & 2-i & -i-1 \\ 0 & 2-i & -1-i \end{pmatrix} \longrightarrow \begin{pmatrix} 1 & -1 & -1 \\ 0 & 2-i & -i-1 \\ 0 & 0 & 0 \end{pmatrix} \begin{matrix} \\ \\ (2-i) \end{matrix} \longrightarrow \begin{pmatrix} 1 & -1 & -1 \\ 0 & 1 & -\frac{1+3i}{5} \\ 0 & 0 & 0 \end{pmatrix} \begin{matrix} A \\ \\ \\ \end{matrix}$

$\longrightarrow \begin{pmatrix} 1 & 0 & -\frac{2}{5}-\frac{1}{5} \\ 0 & 1 & -\frac{1+3i}{5} \\ 0 & 0 & 0 \end{pmatrix} \Rightarrow \begin{matrix} x = \frac{2+i}{5} z \\ y = \frac{1+3i}{5} z \end{matrix} \Leftrightarrow \begin{pmatrix} x \\ y \\ z \end{pmatrix} = \frac{z}{5} \begin{pmatrix} 2+i \\ 1+3i \\ 5 \end{pmatrix}$

$w$.

Uma base é $\mathcal{B} = \{w\}$

$X' = AX$ sabe $\{\varphi_1(t) = e^{2t}v,\ \varphi_2(t) = Re(e^{(2+i)t}w),$

$\varphi_3(t) = Im(e^{(2+i)t}w)\}$

$e^{(2+i)t}w = e^{2t}e^{it}\begin{pmatrix} 2+i \\ 1+3i \\ 5 \end{pmatrix} = e^{2t}\begin{pmatrix} 2\cos t - \sin t \\ \cos t - 3\sin t \\ 5\cos t \end{pmatrix} + i\,e^{2t}\begin{pmatrix} \cos t + 2\sin t \\ 3\cos t + \sin t \\ 5\sin t \end{pmatrix}$

$\varphi_2(t)$ $\qquad$ $\varphi_3(t)$

$\Rightarrow$ Sol geral sabe $\begin{pmatrix} x_1(t) \\ x_2(t) \\ x_3(t) \end{pmatrix} = c_1 e^{2t}\begin{pmatrix} 1 \\ 0 \\ 1 \end{pmatrix} + c_2 e^{2t}\begin{pmatrix} 2\cos t - \sin t \\ \cos t - 3\sin t \\ 5\cos t \end{pmatrix} + c_3 e^{2t}\begin{pmatrix} \cos t + 2\sin t \\ 3\cos t + \sin t \\ 5\sin t \end{pmatrix}$

com $c_1, c_2, c_3 \in \mathbb{R}$

II) Sistema numa base a cuja matriz seja diagonalizável

pasta $\mathcal{C}$

Ex) Só se dá solução geral a este sistema

$\begin{cases} x_1' = x_2 - 5\cos t \\ x_2' = 2x_1 + x_2 \end{cases}, \ t \in \mathbb{R}$

(sc 1) Temos $A = \begin{pmatrix} 0 & 1 \\ 2 & 1 \end{pmatrix}$ e $b(t) = \begin{pmatrix} -5\cos t \\ 0 \end{pmatrix}$

passo 1) Calc valores e vet próprios

$p_A(\lambda) = \det A - \lambda I_2 = \begin{vmatrix} -\lambda & 1 \\ 2 & 1-\lambda \end{vmatrix} = -\lambda(1-\lambda) - 2 =$

$= -\lambda + \lambda^2 - 2 =$

$= \lambda^2 - \lambda - 2 = (\lambda-2)(\lambda+1)$

passo 2) Estudar se A diag.

$\Rightarrow \lambda_1 = 2$

$\lambda_2 = -1$ com multiplic $m_1 = m_2 = 1 \Rightarrow$ A diagonalizável

$\boxed{\lambda_1 = 2}$ Quer $v \in (a,b) \in \mathbb{R}^2$

$Av = \lambda v$

$\begin{pmatrix} 0 & 1 \\ 2 & 1 \end{pmatrix}\begin{pmatrix} a \\ b \end{pmatrix} = 2\begin{pmatrix} a \\ b \end{pmatrix} \Rightarrow \begin{cases} b = 2a \\ 2a + b = 2b \end{cases}$ $\boxed{2a = b}$

Fix $\alpha = a$, $\alpha \in \mathbb{R}$

$\Rightarrow V_{\lambda=2} = \{(\alpha, 2\alpha), \alpha \in \mathbb{R}\} = Sp\{(1,2)\}$

$v$

passo 3) Estabelecer $\varphi$

NU $\boxed{\lambda = -1} \Rightarrow \begin{cases} b = -a \\ 2a + b = -b \end{cases} \Rightarrow 2a = 2b$

$\Downarrow$

Uma base é $\mathcal{B} = \{v\}$ é base de $Sp$ vet $V_{\lambda=2}$

$\Rightarrow \varphi_1 = e^{\lambda_1 t}v = e^{2t}\begin{pmatrix} 1 \\ 2 \end{pmatrix}$

$\lambda_2 = -1$ $\quad b = -a$ (vezi pag ant) $\Rightarrow$

Ex $b = \underset{\alpha \in \mathbb{R}}{\alpha} \Rightarrow V_{\lambda=-1} = \{(-\alpha, \alpha), \alpha \in \mathbb{R}\} = Sp\{(-1,1)\}$

$\{v_2\}$ este o bază pt sp vect real $V_{\lambda_2=-1}$

$\varphi_2(t) = e^{\lambda t} v_2 = e^{-t}\begin{pmatrix} -1 \\ 1 \end{pmatrix}$

5) Stabilim sol fundamentală $X_n$

Deci $\{\varphi_1(t), \varphi_2(t)\}$ este o bază pt sp vect real al ec.

$X_{\varphi}(t) = c_1\varphi_1(t) + c_2\varphi_2(t) = c_1 e^{2t}\begin{pmatrix} 1 \\ 2 \end{pmatrix} + c_2 e^{-t}\begin{pmatrix} -1 \\ 1 \end{pmatrix}$

c) Caut $X_p$.

! $X_p$ are forma

$X_p(t) = C_1(t)\varphi_1(t) + C_2(t)\varphi_2(t)$, unde $C_1', C_2'$ verif

$C_1'\varphi_1(t) + C_2'\varphi_2(t) = f(t)$

$\Leftrightarrow C_1' e^{2t}\begin{pmatrix} 1 \\ 2 \end{pmatrix} + C_2' e^{-t}\begin{pmatrix} -1 \\ 1 \end{pmatrix} = \begin{pmatrix} -5\cos t \\ 0 \end{pmatrix}$

$\Leftrightarrow \begin{cases} C_1' e^{2t} - C_2' e^{-t} = -5\cos t \\ 2C_1' e^{2t} + C_2' e^{-t} = 0 \end{cases}$ $\quad (+)$

$3C_1' e^{2t} = -5\cos t$

$C_1' = -\frac{5}{3}\cos t \cdot e^{-2t}$

$C_1(t) = -\frac{e^{-2t}}{3}(\sin t - 2\cos t) + A_1$

$C_2' = -2e^{3t} \cdot C_1'(t) = \frac{10}{3} e^t \cos t$

$C_2(t) = \frac{5}{3} e^t(\cos t + \sin t) + A_2$

Pt $A_1 = A_2 = 0$

$\Rightarrow X_p = -\frac{e^{-2t}}{3}(\sin t - 2\cos t)\, e^{2t}\begin{pmatrix} 1 \\ 2 \end{pmatrix} + \frac{5}{3} e^t(\cos t + \sin t) e^{-t}\begin{pmatrix} -1 \\ 1 \end{pmatrix}$

$= \frac{1}{3}\begin{pmatrix} -6\sin t - 3\cos t \\ 3\sin t + 9\cos t \end{pmatrix} = \begin{pmatrix} -\cos t - 2\sin t \\ 3\cos t + \sin t \end{pmatrix}$

$\Rightarrow X = \begin{pmatrix} x_1(t) \\ x_2(t) \end{pmatrix} = c_1 e^{2t}\begin{pmatrix} 1 \\ 2 \end{pmatrix} + c_2 e^{-t}\begin{pmatrix} -1 \\ 1 \end{pmatrix} + \begin{pmatrix} -\cos t - 2\sin t \\ 3\cos t + \sin t \end{pmatrix}$

- Exponențiala unei matrici

$e^A = \sum_{n=0}^{\infty} \frac{1}{n!} A^n = I_n + \frac{1}{1!}A + \frac{1}{2!}A^2 + \dots + \frac{1}{n!}A^n + \dots$

Obs 1) $e^{At} = \sum_{n=0}^{\infty} \frac{t^n}{n!} A^n$

2) $e^{At}$ este matrice fund. de soluții pt sist $X' = AX \Rightarrow \boxed{X(t) = e^{At} \cdot C}$, $C = \begin{pmatrix} c_1 \\ c_2 \\ c_n \end{pmatrix}$

Calculul ei

$e^{At} = \alpha_{n-1} A^{n-1} t^{n-1} + \alpha_{n-2} A^{n-2} t^{n-2} + \dots + \alpha_1 At + \alpha_0 I_n$

metoda $r(\lambda) = \alpha_{n-1}\lambda^{n-1} + \dots + \alpha_1\lambda + \alpha_0$

$\mu \in \sigma(At)$ cu mult $m \geq 1$ ac lui

$r(\mu) = e^{\mu}$

$r'(\lambda)\big|_{\lambda=\mu} = e^{\mu}$

$r^{(m-1)}(\lambda)\big|_{\lambda=\mu} = e^{\mu}$.

Ex Sol o seg problema Cauchy $X' = AX$

$$A = \begin{pmatrix} 0 & 1 & 0 \\ 0 & 0 & 1 \\ 1 & -3 & 3 \end{pmatrix} \quad \text{si} \quad X(0) = \begin{pmatrix} -2 \\ 0 \\ 1 \end{pmatrix}$$

R Fie $X(t) = \begin{pmatrix} x_1(t) \\ x_2(t) \\ x_3(t) \end{pmatrix} \Rightarrow$ Sol gen $\boxed{X(t) = e^{At} \begin{pmatrix} c_1 \\ c_2 \\ c_3 \end{pmatrix}}$ $t=0$ $\Rightarrow$

$\Rightarrow X(t) = e^{At} X(0)$

Pasaño a fol exponencial

• alg val propios

$$P_A(\lambda) = \det A - \lambda I_3 = \begin{vmatrix} -\lambda & 1 & 0 \\ 0 & -\lambda & 1 \\ 1 & -3 & 3-\lambda \end{vmatrix} = \lambda^2(3-\lambda) + 1 - 3\lambda =$$

$$= -(\lambda^3 - 3\lambda^2 + 3\lambda - 1) = -(\lambda-1)^3$$

$\Rightarrow \sigma(A) = \{1\}$ cu mult $m = 3 \Rightarrow \sigma(At) = \{t\}$

si $m_{\mu=t} = 3$

• Scriu forma gen $e^{At}$

$\Rightarrow e^{At} = \alpha_2 A^2 t^2 + \alpha_1 At + \alpha_0 I_3$

$(n=3)$

• Fac calcem de unde scot $\alpha_0, \alpha_1, \alpha_2$

Fie $r(\lambda) \overset{not}{=} \alpha_2 \lambda^2 + \alpha_1 \lambda + \alpha_0$

$\Rightarrow r'(\lambda) = 2\lambda\alpha_2 + \alpha_1$

$r''(\lambda) = 2\alpha_2$

Rezolv urm sistem pt a det functiile $\alpha_0, \alpha_1, \alpha_2$

$$\begin{cases} r(t) = e^t \\ r'(\lambda)\big|_{\lambda=t} = e^t \\ r''(\lambda)\big|_{\lambda=t} = e^t \end{cases} \Leftrightarrow \begin{cases} \alpha_2 t^2 + \alpha_1 t + \alpha_0 = e^t \\ 2\alpha_2 t + \alpha_1 = e^t \\ 2\alpha_2 = e^t \end{cases} \Leftrightarrow$$

$\Leftrightarrow \alpha_2 = \frac{e^t}{2}$

$e^t t + \alpha_1 = e^t \Rightarrow \alpha_1 = e^t(1-t)$

$$\alpha_0 = e^t - \frac{e^t}{2} t^2 - e^t(1-t)t =$$

$$= e^t\left(1 - \frac{t^2}{2} - t + t^2\right) = e^t\left(\frac{t^2}{2} - t + 1\right) =$$

$$= \frac{e^t}{2}(t^2 - 2t + 2)$$

Acum, inlocuiesc in formula

$$e^{At} = \alpha_2 A^2 t^2 + \alpha_1 At + \alpha_0$$

$$e^{At} = \frac{e^t t^2}{2}\begin{pmatrix} 0 & 0 & 1 \\ 1 & -3 & 3 \\ 3 & -8 & 6 \end{pmatrix} + \frac{1}{2}e^t(2t - 2t^2)\begin{pmatrix} 0 & 1 & 0 \\ 0 & 0 & 1 \\ 1 & -3 & 3 \end{pmatrix} + \frac{e^t}{2}(t^2 - 2t + 2)\begin{pmatrix} 1 & 0 & 0 \\ 0 & 1 & 0 \\ 0 & 0 & 1 \end{pmatrix}$$

$$e^{At} = \frac{e^t}{2}\begin{pmatrix} t^2 - 2t^2 + 2t + 2 & 2t - 2t^2 & t^2 \\ t^2 & -2t^2 - 2t + 2 & t^2 + 2t \\ t^2 + 2t & -2t^2 - 6t & t^2 + 4t + 2 \end{pmatrix}$$

Deci $\begin{pmatrix} x_1(t) \\ x_2(t) \\ x_3(t) \end{pmatrix} = e^{At} X(0) = e^{At}\begin{pmatrix} -2 \\ 0 \\ 1 \end{pmatrix}$

Ex 4 De rezolvat cat mai folosind metoda derivării și eliminării (să reducem la o ec dif liniară de ordinul II)

$$\begin{cases} x_1' = x_2 - 5\cos t \\ x_2' = 2x_1 + x_2 \end{cases} \overset{\text{scot o var}}{\Longrightarrow} x_1 = \frac{x_2' - x_2}{2}$$

$$\Rightarrow x_1'' = \frac{x_2'' - x_2'}{2}$$

Inlocuiesc in cealaltă ec $x_2'' - x_2' = 2x_2 - 10\cos t$

$\Rightarrow x_2'' - x_2' - 2x_2 = -10\cos t$

$\lambda^2 - \lambda - 2 = 0$

$\Delta = 9 \Rightarrow \lambda_{1,2} = \frac{1 \pm 3}{2}$ $\begin{cases} \lambda_1 = 2 \text{ , } m_{a_1} = 1 \rightarrow e^{2t} \\ \lambda_2 = -1 \text{ , } m_{a_2} = 1 \rightarrow e^{-t} \end{cases}$ sol fund

$x_2(t) = c_1 e^{2t} + c_2 e^{-t}$ , $c_1, c_2 \in \mathbb{R}$

· Agora $x_p$ (observando coeficientes)

$e^{at}(P_m(x)\cos bt + Q_m(t)\sin bt) = -10\cos t$

$\begin{cases} a = 0 \\ b = 1 \end{cases} \Rightarrow a + bi = i$ não é solução $\Rightarrow s = 0$.

$(\sin x)' = \cos x$
$(\cos x)' = -\sin x$

$\left.\begin{array}{l} P_m(t) = -10 \\ Q_m(t) \overset{\text{abuso}}{=} 0 \end{array}\right\} \Rightarrow N = 0 = \max(m, w)$

$t^s e^{at}(\tilde{P}_0(t)\cos bt + \tilde{Q}_0(t)\sin bt) = x_{2p}$

$x_{2p} = \alpha\cos t + \beta\sin t$

$(x_2)'_p = -\alpha\sin t + \beta\cos t$

$(x_2)''_p = -\alpha\cos t - \beta\sin t$

Substituir na eq diff

$x_2'' - x_2' - 2x_2 = -10\cos t$

$-\alpha\cos t - \beta\sin t + \alpha\sin t - \beta\cos t - 2\alpha\cos t - 2\beta\sin t = -10\cos t$

$-\beta\cos t - 3\alpha\cos t - 3\beta\sin t + \alpha\sin t = -10\cos t$

$\begin{cases} -\beta - 3\alpha = -10 \\ -3\beta + \alpha = 0 \end{cases} \Rightarrow \begin{array}{l} 10\beta = 10 \\ \alpha = 3\beta \end{array} \Rightarrow \begin{array}{l} \beta = 1 \\ \alpha = 3 \end{array}$

Logo $x_p^2 = 3\cos t + \sin t$

$x_2(t) = x_{h_2}(t) + x_{p_2}(t) =$

$= c_1 e^{2t} + c_2 e^{-t} + 3\cos t + \sin t$

$\Rightarrow x_1(t) = \frac{x_2'(t) - x_2(t)}{2} =$

$= \frac{1}{2}\left(2c_1 e^{2t} - c_2 e^{-t} - 3\sin t + \cos t - c_1 e^{2t} - c_2 e^{-t} - 3\cos t - \sin t\right)$ $c_1, c_2 \in \mathbb{R}$

COROLAR III) Sistema diferencial linear homogéneo de ordem 1 com coeficientes constantes

Solução geral do SDL acima

$X(t) = e^{At}\left(\begin{pmatrix} c_1 \\ c_n \end{pmatrix} + \int e^{-At} b(t)\, dt\right)$

Dado $X(t_0) = \begin{pmatrix} \alpha_1 \\ \alpha_n \end{pmatrix}$ at

$$X(t) = e^{A(t-t_0)}\begin{pmatrix} \alpha_1 \\ \alpha_2 \\ \alpha_n \end{pmatrix} + \int_{t_0}^{t} e^{A(t-s)} b(s)\, ds$$

(ex) Fe SDL $\begin{cases} x' = 3x - y + e^{3t} \\ y' = 5x - 3y - e^{2t} \end{cases}$

a) Determine a matriz associada ao SDL, diga se é diagonalizável

b) Determine a solução do problema

a) Folosesc exponentiala

$$A = \begin{pmatrix} 3 & -1 \\ 5 & -3 \end{pmatrix}$$

1. Calculez valorile proprii

$$P_A(\lambda) = \det A - \lambda I_2 = \begin{vmatrix} 3-\lambda & -1 \\ 5 & -3-\lambda \end{vmatrix} = (3-\lambda)(-3-\lambda) + 5 =$$
$$= -9 - 3\lambda + 3\lambda + \lambda^2 + 5 =$$
$$= \lambda^2 - 4 = \lambda^2 - 4$$

$\Rightarrow \lambda_{1,2} = \pm 2$ cu $m_1 = m_2 = 1$

$\Rightarrow V(A) = \{\underbrace{-2}_{\mu_1}, \underbrace{2}_{\mu_2}\} \Rightarrow V(At) = \{\underbrace{-2t}_{\mu_1}, \underbrace{2t}_{\mu_2}\}$

2) Determinam $e^{At}$

$e^{At} \overset{(n=2)}{=} \alpha_1 A t + \alpha_0 I_2 \rightarrow$ forma generala $e^{At}$

$r(\lambda) \overset{not}{=} \alpha_1 \lambda + \alpha_0 \rightarrow$ restul $r(\lambda)$

$r(\mu_1) = e^{\mu_1} = e^{-2t} \qquad r'(\lambda) = \alpha_1$

$r(\mu_2) = e^{\mu_2} = e^{2t}$

?

$$\left.\begin{cases} -2t\,\alpha_1 + \alpha_0 = e^{-2t} \\ 2t\,\alpha_1 + \alpha_0 = e^{2t} \end{cases}\right\} \text{Fac sistem in } \alpha_1 \text{ si } \alpha_0$$

$$\alpha_0 = \frac{e^{2t} + e^{-2t}}{2}$$

$$\Rightarrow \alpha_1 = \frac{1}{2t}\left(e^{2t} - \frac{e^{2t} + e^{-2t}}{2}\right) = \frac{1}{4t}\left(e^{2t} - e^{-2t}\right)$$

$$e^{At} = \frac{1}{4}\left(e^{2t} - e^{-2t}\right)\begin{pmatrix} 3 & -1 \\ 5 & -3 \end{pmatrix} + \frac{e^{2t} + e^{-2t}}{2}\begin{pmatrix} 1 & 0 \\ 0 & 1 \end{pmatrix}$$

$$= \frac{1}{4}\begin{pmatrix} 3e^{2t} - e^{-2t} & -e^{2t} + e^{-2t} \\ 5e^{2t} - 5e^{-2t} & -e^{2t} + 3e^{-2t} \end{pmatrix}$$

b) Folosesc formula

$$\int e^{-At} b(t)\,dt = \frac{1}{4}\int \begin{pmatrix} 3e^{-2t} - e^{2t} & e^{2t} - e^{-2t} \\ 5e^{-2t} - 5e^{2t} & -e^{2t} + 3e^{-2t} \end{pmatrix}\begin{pmatrix} e^{2t} \\ -e^{-2t} \end{pmatrix} dt$$

Sol gen este

$$\begin{pmatrix} x(t) \\ y(t) \end{pmatrix} = e^{At}\left(\begin{pmatrix} c_1 \\ c_2 \end{pmatrix} + \int e^{-At} b(t)\,dt\right) \quad \text{cu } c_1, c_2 \in \mathbb{R}$$

b) Folosesc met directa si eliminarii

$$\begin{cases} x' = 3x - y + e^{2t} \\ y' = 5x - 3y - e^{2t} \end{cases} \overset{\text{derivez}}{\Leftrightarrow} \boxed{x'' = 3x' - y' + 3e^{2t}}$$

$\Downarrow$

$y' = -x'' + 3x' + 3e^{2t}$

Din prima ec, il scot pe $y$: $\boxed{y = -x' + e^{2t} + 3x}$

$$\Rightarrow y' = 5x + 3x' - 3e^{2t} - 9x - e^{2t}$$
$$y' = 3x' - 4x - 3e^{2t} - e^{2t}$$

Acum inlocuiesc in $x''$

$$x'' - 3x' \overset{=}{+} y' = 3e^{2t}$$
$$x'' - 3x' + 3x' - 4x - 3e^{2t} - e^{2t} = 3e^{2t}$$
$$\Downarrow$$
$$x'' - 4x = 6e^{2t} + e^{2t}$$

Calc ec caract si $\det X_h$

$\lambda^2 - 4 = 0 \Rightarrow \lambda_{1,2} = \pm 2$ cu $m_1 = m_2 = 1$

$\Rightarrow X_1(t) = e^{2t}$
$X_2(t) = e^{-2t}$ $\rangle$ sol fundamentala

$\rightsquigarrow x_h = c_1 e^{2t} + c_2 e^{-2t}$, $c_1, c_2 \in \mathbb{R}$

Cont soluções particulares

$(x_p^1)$ $x'' - 4x = 6e^{3t}$ $(x_p^2)$ $x'' - 4x = +e^{2t}$

$y(t) = e^{\alpha t}(P_m(t)\cos\beta t + Q_m(t)\sin\beta t)$

$\left.\begin{array}{l} a = 3 \\ b = 0 \end{array}\right| \Rightarrow a + ib = 3$ não é sol $\Rightarrow \boxed{S = 0}$

$x_p^1 = t^S \cdot e^{\alpha t}(\tilde{P}_N \cos\beta t + \tilde{Q}_N \sin\beta t)$

$\left.\begin{array}{l} P_m(t) = 6 \\ Q_m(t) = 0 \end{array}\right\} \Rightarrow \boxed{N = 0}$

Para $x_p^1 = \alpha e^{3t} \cdot (\quad)$

$(x_p^1)' = 3\alpha e^{3t}$

$(x_p^1)'' = 9\alpha e^{3t}$

$9\alpha e^{3t} - 4\alpha e^{3t} = 6e^{3t}$

$\Rightarrow 5\alpha = 6 \Rightarrow \alpha = \frac{6}{5} \Rightarrow x_p^1 = \frac{6}{5} e^{3t}$

$(x_p^2)$ $\left.\begin{array}{l} a = 2 \\ b = 0 \end{array}\right| \Rightarrow a + ib = 2$ solução pt ec característica

$\Rightarrow \boxed{S = 1}$

$\left.\begin{array}{l} P_m(t) = 1 \\ Q_m(t) = 0 \end{array}\right\} \Rightarrow N = 0$

Para $x_p^2(t) = t \cdot e^{2t} \cdot p$

$(x_p^2)' = 1 \cdot p \cdot e^{2t} + 2te^{2t}p = e^{2t}p(1+2t)$

$(x_p^2)'' = 4pe^{2t}(1+2t) + 4te^{2t}p$

$\Downarrow$

$(x_p^2)'' = pe^{2t}(4t+4)$

$pe^{2t}(4t+4) - 4te^{2t}p = e^{2t}$

$\Downarrow$

$4p = 1 \Rightarrow \boxed{p = \frac{1}{4}}$

$(x_p^2)(t) = \frac{1}{4} te^{2t}$

Logo $x(t) = c_1 e^{2t} + c_2 e^{-2t} - 2e^{3t} + \frac{1}{4} te^{2t}$

$\forall$ indicação após $\forall$ $y(t)$

ex2 Falsamente falta-lhe $A = \begin{pmatrix} 1 & 0 & 1 \\ 0 & 1 & 1 \\ 0 & 0 & 0 \end{pmatrix}$ diagonalizável ou não se não $e^{At}$

Sol $\Gamma(A) = \{0, 1\}$ (1 e 2 mult)

$\boxed{\lambda = 1}$ Cont $v \in (a,b,c)$ vetor próprio associado

$Av^t = \lambda v^t$

$\Downarrow$

$\begin{pmatrix} 1 & 0 & 1 \\ 0 & 1 & 1 \\ 0 & 0 & 0 \end{pmatrix}\begin{pmatrix} a \\ b \\ c \end{pmatrix} = \begin{pmatrix} a \\ b \\ c \end{pmatrix} \Rightarrow \begin{cases} a + c = a \\ b + c = b \\ c = 0 \end{cases}$

Fa $a = \alpha$ e $b = \beta$

$\Rightarrow V_{\lambda=1} = \{(\alpha, \beta, 0),\ \alpha, \beta \in \mathbb{R}\} = Sp\{\underbrace{(1,0,0)}_{v_1}, \underbrace{(0,1,0)}_{v_2}\}$

$\{v_1, v_2\}$ base pt $V_{\lambda=1}$

$\boxed{\lambda = 0} \Rightarrow \begin{cases} a + c = 0 \\ b + c = 0 \end{cases} \Rightarrow \begin{array}{l} -a - c = 0 \\ b + c = 0 \end{array} \Rightarrow \boxed{b = -c}$

$b - a = 0 \Rightarrow \underline{a = b}$

Fa $a = \alpha$

$V_{\lambda=0} = \{(\alpha, \alpha, -\alpha),\ \alpha \in \mathbb{R}\} = Sp\{(1,1,-1)\}$

$\Rightarrow \{v_3\}$ base pt $V_{\lambda=0}$

A diagonalizável $\Leftrightarrow \exists\, T \in M_3(\mathbb{R})$ a.t. $T^{-1}AT = D$

$T = (v_1|v_2|v_3) = \begin{pmatrix} 1 & 0 & 1 \\ 0 & 1 & 1 \\ 0 & 0 & -1 \end{pmatrix}$ ou seja $T^{-1}AT = \begin{pmatrix} 1 & 0 & 0 \\ 0 & 1 & 0 \\ 0 & 0 & -1 \end{pmatrix} = D$.

$\Leftrightarrow A = TDT^{-1} \Rightarrow A^n = TD^nT^{-1}$

Após $T^{-1}$

$(T|I_3) = \left(\begin{array}{ccc|ccc} 1 & 0 & 1 & 1 & 0 & 0 \\ 0 & 1 & 1 & 0 & 1 & 0 \\ 0 & 0 & -1 & 0 & 0 & 1 \end{array}\right) \xrightarrow{L_1 \to L_1 + L_3} \left(\begin{array}{ccc|ccc} 1 & 0 & 0 & 1 & 0 & 1 \\ 0 & 1 & 0 & 0 & 1 & 1 \\ 0 & 0 & -1 & 0 & 0 & 1 \end{array}\right)$

$\longrightarrow \left(\begin{array}{ccc|ccc} 1 & 0 & 0 & 1 & 0 & 1 \\ 0 & 1 & 0 & 0 & 1 & 1 \\ 0 & 0 & 1 & 0 & 0 & -1 \end{array}\right) \Rightarrow T^{-1} = \begin{pmatrix} 1 & 0 & 1 \\ 0 & 1 & 1 \\ 0 & 0 & -1 \end{pmatrix}$

$I_3$

$A^n = TD^nT^{-1} = \begin{pmatrix} 1 & 0 & 1 \\ 0 & 1 & 1 \\ 0 & 0 & -1 \end{pmatrix}\begin{pmatrix} 1 & 0 & 0 \\ 0 & 1 & 0 \\ 0 & 0 & 0 \end{pmatrix}\begin{pmatrix} 1 & 0 & 1 \\ 0 & 1 & 1 \\ 0 & 0 & -1 \end{pmatrix} = \ldots = A$

Dado $e^{At} = I_3 + \frac{1}{1!}At + \frac{1}{2!}(At)^2 + \ldots + \frac{1}{n!}(At)^n + \ldots$

$\Rightarrow e^{At} = I_3 + \underbrace{\left(\frac{1}{1!}t + \frac{t^2}{2!} + \ldots + \frac{t^n}{n!} + \ldots\right)}_{e^t - 1} A$

$= \begin{pmatrix} 1 & 0 & 0 \\ 0 & 1 & 0 \\ 0 & 0 & 1 \end{pmatrix} + (e^t - 1)\begin{pmatrix} 1 & 0 & 1 \\ 0 & 1 & 1 \\ 0 & 0 & 0 \end{pmatrix} = \begin{pmatrix} e^t & 0 & e^t - 1 \\ 0 & e^t & e^t - 1 \\ 0 & 0 & 1 \end{pmatrix}$

**ex3** Só se dá solução problema Cauchy

$x^3y''' + 3x^2y'' + 2xy' = 2x$

$y(1) = 0$, $y'(1) = 1$ e $y''(1) = -1$.

Obvio é um a equação EULER

Faz-se mudança de variável $x = e^t$ e $z(t) = y(e^t)$

Dado $\begin{cases} y'(t) = e^{-t}\, z' \\ y''(t) = e^{-2t}(z'' - z') \\ y'''(t) = e^{-3t}(z''' - 3z'' + 2z') \end{cases}$

Assim substituindo na equação

$z''' - 3z'' + 2z' + 3z'' - 3z' + 2z' = 2e^t$

$\Downarrow$

$z''' + z' = 2e^t$

$\Downarrow$

$\lambda^3 + \lambda = 0$

$\Downarrow$

$\lambda(\lambda^2 + 1) = 0 \Rightarrow \lambda = 0$ ou $m_1 = 1$

$\lambda_{2,3} = \pm i$ ou $m_{2,3} = 1$

$\Delta = -4$

$\lambda_{1,2} = \frac{\pm 2i}{2}$

Assim $z_1(t) = e^{0t} = 1$

$z_2(t) = e^{it}$

$z_3(t) = e^{-it}$

Dado $z_h(t) = c_1 + c_2\underbrace{e^{it}}_{\cos t} + c_3\underbrace{e^{-it}}_{\operatorname{sen} t}$

$e^{it} = \cos t + i \operatorname{sen} t$

$e^{-it} = \ldots$

$\alpha = 1$, $\beta = 0$ $\Rightarrow \alpha + \beta i = i$ não é raiz $\Rightarrow S = 0$.

$f(x) = e^{\alpha x}(P_m(x)\cos bx + Q_m(x)\operatorname{sen} bx)$

$P_m(t) = 2$

$Q_m(t) = 0$ $\Rightarrow N = 0$

Dado $z_p = \alpha e^t$

$\cos 0 = 1$

$z_p(t) = (\alpha e^t)' = \alpha e^t$

$z_p''(t) = z_p'''(t) = \alpha e^t$

Deci $2\alpha = 2 \Rightarrow \boxed{\alpha = 1} \Rightarrow \boxed{z_p(t) = e^t}$

$z(t) = c_1 + c_2 \cos t + c_3 \sin t + e^t$

$\Rightarrow y(x) = c_1 + c_2 \cos(\ln x) + c_3 \sin(\ln x) + x$

$y(1) = 0 \Rightarrow c_1 + c_2 + 1 = 0 \Rightarrow \boxed{c_1 + c_2 = -1}$

$y'(1) = 1 \Rightarrow$

$y'(x) = c_2 \cdot \left(-\frac{\sin(\ln x)}{x}\right) + c_3 \frac{\cos(\ln x)}{x} + 1$

$\Downarrow y'(1) = 1$

$1 = 1 + c_3 \Rightarrow \boxed{c_3 = 0}$

$y''(x) = -c_2 \left(-\frac{1}{x^2}\sin(\ln x) + \frac{1}{x}\frac{\cos(\ln x)}{x}\right)$

$\Downarrow y''(1) = -1$

$-c_2 = -1 \Rightarrow \boxed{c_2 = 1} \Rightarrow \boxed{c_1 = -2}$

În final $y(x) = -2 + \cos(\ln x) + x$

$p_{U^\perp} v = \frac{\langle v, w_1 \rangle}{\langle w_1, w_1 \rangle} \cdot w_1 + \frac{\langle v, w_2 \rangle}{\langle w_2, w_2 \rangle} w_2$

**exercitiu** Să se afle termenul general al șirului $(x_n)_{n \geq 1}$ definit prin

$x_{n+2} = 15 x_{n+1} - 56 x_n$, știind că $x_1 = x_2 = 1$

R Fie $X_n \overset{not}{=} \begin{pmatrix} x_n \\ x_{n-1} \end{pmatrix} \Rightarrow X_{n+2} = \begin{pmatrix} x_{n+2} \\ x_{n+1} \end{pmatrix} = \begin{pmatrix} 15 & -56 \\ 1 & 0 \end{pmatrix} \begin{pmatrix} x_{n+1} \\ x_n \end{pmatrix}$

$X_{n+2} = A X_{n+1} \Rightarrow X_n = A X_{n-1} = A^2 X_{n-2} = \ldots = A^{n-2} X_2 = A^{n-2} \begin{pmatrix} x_2 \\ x_1 \end{pmatrix} = \begin{pmatrix} 1 \\ 1 \end{pmatrix}$

$P_A(\lambda) = \begin{vmatrix} 15-\lambda & -56 \\ 1 & -\lambda \end{vmatrix} = \lambda^2 - 15\lambda + 56 = (\lambda - 7)(\lambda - 8)$

$\Rightarrow V(A) = \{7, 8\}$ val proprii simple $\Rightarrow A$ diagonalizabilă

$(\lambda = 7)$ Caut $v \in (a, b)$ nenul pentru care

$Av^t = \lambda v^t$

$\begin{pmatrix} 15 & -56 \\ 1 & 0 \end{pmatrix} \begin{pmatrix} a \\ b \end{pmatrix} = 7 \begin{pmatrix} a \\ b \end{pmatrix}$

$15a - 56b = 7a$

$a = 7b \Rightarrow b = \frac{a}{7}$

Fie $b = \alpha \Rightarrow V_{\lambda=7} = \{(7\alpha, \alpha) \mid \alpha \in \mathbb{R}\} = \operatorname{Sp}\{(7, 1)\}$, $v_1$

$\{v_1\}$ bază pt $V_{\lambda=7}$

$(\lambda = 8)$ $15a - 56b = 8a$

$a = 8b \Rightarrow V_{\lambda=8} = \{(8\alpha, \alpha) \mid \alpha \in \mathbb{R}\} = \operatorname{Sp}\{(8, 1)\}$, $v_2$

$\{v_2\}$ bază pt $V_{\lambda=8}$

$T \overset{not}{=} (v_1 | v_2) = \begin{pmatrix} 7 & 8 \\ 1 & 1 \end{pmatrix}$ are loc $T^{-1}AT = D = \begin{pmatrix} 7 & 0 \\ 0 & 8 \end{pmatrix}$

$\Leftrightarrow A = TDT^{-1}$

ind $\Rightarrow A^n = TD^nT^{-1}$ $D^n = \begin{pmatrix} 7^n & 0 \\ 0 & 8^n \end{pmatrix}$

$T^{-1} = \cancel{\begin{pmatrix} 1 & -8 \\ -1 & 7 \end{pmatrix}}$ $T^{-1} = \begin{pmatrix} -1 & 8 \\ 1 & -7 \end{pmatrix}$

$\Rightarrow A^n = \begin{pmatrix} 7 & 8 \\ 1 & 1 \end{pmatrix}\begin{pmatrix} 7^n & 0 \\ 0 & 8^n \end{pmatrix}\begin{pmatrix} -1 & 8 \\ 1 & -7 \end{pmatrix} =$

$= \begin{pmatrix} 7^{n+1} & 8^{n+1} \\ 7^n & 8^n \end{pmatrix}\begin{pmatrix} -1 & 8 \\ 1 & -7 \end{pmatrix} = \begin{pmatrix} -7^{n+1} + 8^{n+1} & 8\cdot 7^{n+1} - 7\cdot 8^{n+1} \\ 8^n - 7^n & 8\cdot 7^n - 7\cdot 8^n \end{pmatrix}$

$n \to n+1$
$n-2 \to n+1-2 = n-1$

Deci $X_n = \begin{pmatrix} x_n \\ x_{n-1} \end{pmatrix} = \underbrace{\begin{pmatrix} 8^{n-1} - 7^{n-1} & 8\cdot 7^{n-1} - 7\cdot 8^{n-1} \\ 8^{n-2} - 7^{n-2} & 8\cdot 7^{n-2} - 7\cdot 8^{n-2} \end{pmatrix}}_{A^{n-2}} \begin{pmatrix} x_1 \\ x_2 \end{pmatrix}$

$\begin{pmatrix} 1 \\ 1 \end{pmatrix}$

$\Rightarrow \boxed{x_n = 7^n - 6\cdot 8^{n-1}}$ $\forall n \geq 1$